THE MASTERMIND COACHING CLASSES

Revision Notes

Class 9 · Social Science

Exploring Society: India and Beyond — Chapters 1 to 8 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Building Blocks in Economics: The Problem of Choice	Scarcity, opportunity cost, PPC, three key questions, economic systems
2	The Price Puzzle: What Drives the Market	Demand, supply, equilibrium and government intervention
3	Shaping the Earth's Surface	Plate tectonics, weathering, erosion, landforms and landform disasters
4	Atmosphere and Climate	Atmosphere layers, weather and climate, monsoon, climate change
5	Early Humans and the Beginning of Civilisation	Early humans, Stone Age to Bronze Age, Harappan origins and river-valley civilisations
6	State and Society up to 1000 CE	Vedic polity, janapadas to empires, administration, society, education, economy
7	Democracy	Evolution, principles, types and challenges of democracy in India
8	Elections	Electoral systems, laws, Election Commission, political parties and election challenges

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CHAPTER 01 · ECONOMICS

Building Blocks in Economics: The Problem of Choice

Money, time, land and even steel are limited, but what people want has no end. This chapter shows how that gap forces every household, firm and government to choose, what each choice costs, and how different economic systems decide the answers.

Big Questions

- What does economics deal with?
- What are the key questions in economics?
- How do different economic systems address these questions?

Everyday Choices, Needs and Wants

Spending pocket money on snacks or saving for shoes, deciding who gets the one library copy of a popular book, a farmer picking a crop, a firm choosing between more workers or more machines, a government choosing between highways and hospitals — all are **economic choices** made by individuals, enterprises and governments.

	Needs	Wants
Meaning	Essentials without which life or work is difficult	Things we would like to have for comfort or pleasure
Examples	Food, water, shelter, medicines	Gadgets, holidays, luxury items
Nature	Limited and fairly stable	**Unlimited** and keep changing (bicycle → motorbike → car)

DEFINITION

Market is a place (physical or virtual, on the internet) where products and services are bought and sold.

Choices and Limited Resources

- **Resources** are the factors used to produce goods and services. They may be natural (water, coal) or human-made (capital, technology).
- Factors of production, learnt in Grade 8, are **land, labour, capital and technology**.
- Both natural and human-made resources are **limited** in quantity, yet each has **many alternative uses**. Steel, for example, can go into aircraft or into refrigerators.
- Because wants are unlimited and resources limited, there is **scarcity**, and scarcity forces **choice**: households and whole economies must use scarce resources in the best possible way to improve quality of life.

▸ Opportunity cost

When one alternative is chosen, the others are given up. The **value of what is given up** is the **opportunity cost** of the choice.

▸ Worked example — the farmer's barley and wheat

A farmer has one plot with limited land, water and labour and can grow barley or wheat. The possible combinations are:

Combination	Barley (kg)	Wheat (kg)
A	0	100
B	25	90
C	50	70
D	75	40
E	100	0

- Plotting barley on the x-axis and wheat on the y-axis gives a **downward-sloping curve**, the **Production Possibility Curve (PPC)**.
- The PPC shows the **different combinations of goods that can be produced using all available resources efficiently**, with no wastage.
- Moving from A to E gives more barley but less wheat. The wheat sacrificed is the **opportunity cost of growing more barley**.
- Every point on the curve is a maximum output, so the PPC helps enterprises and governments plan and decide.

> **REMEMBER**
> Opportunity cost can be read off the table: from A to B, 25 kg barley costs 10 kg wheat; from B to C, 25 kg more barley costs 20 kg wheat; from C to D it costs 30 kg wheat, and from D to E it costs 40 kg wheat. Giving up more and more wheat for each extra 25 kg of barley shows the trade-off clearly.

Time is also a scarce resource: how you spend an evening is a choice with an opportunity cost.

What Does Economics Deal With?

- The word **economics** comes from the Greek *oikonomia*, made of *oikos* (household) and *nemein* (management). Economics therefore began as **household management**; the same idea applies to nations that must plan limited resources.
- Economics studies **how choices are made to use limited resources to satisfy needs and wants**.
- It explains how **economic entities** (consumers, producers, government, enterprises, financial institutions) interact — how people work and earn wages, how wealth and resources are shared, how prices are set in the market, how education and technology drive investment, and how government policy and trade affect prices and jobs.

Good decisions need **data and analysis, not guesswork**. Families divide income among essential items (food, medicines, school supplies), non-essential items (jewellery, entertainment, restaurants) and savings. Governments use tax revenue to plan spending on infrastructure and welfare. Enterprises study market trends and new innovations to serve customers and increase profit. Economists weigh the alternatives, their opportunity costs and likely outcomes, using government reports such as economic surveys and company financial statements.

Area where economists work	What they do
Policy-making	Guide governments on taxation and welfare spending

Area where economists work	What they do
Business consulting	Help firms plan growth and improve efficiency
Finance	Advise investors on where to invest
Research and education	Study economic trends and teach others

NOTE

The **Economic Survey of India** is an annual document of the Ministry of Finance, presented in Parliament **before the Union Budget**. It reviews the previous year's performance of agriculture, industry, services, employment, inflation, education, health and infrastructure, discusses future challenges, and works as a blueprint for the Budget. Terms to know: *survey* (systematic collection and analysis of data), *data* (facts and statistics) and *policy* (a course of action adopted by an organisation or individual).

Key Questions in Economics

The mismatch between **unlimited wants and limited resources** creates scarcity and choices, which are summed up in three questions: **what to produce, how to produce and for whom to produce**.

▸ What to produce (and in what quantity)

- The economy must decide which goods and services, and how much of each, to produce over a period.
- Example: water-intensive sugarcane and paddy versus drought-resistant millets and pulses. Sugarcane gives high profit and supports the sugar industry; millets and pulses save water, improve soil health and promote sustainable farming. The opportunity cost of sugarcane is the saving in water and soil health forgone.
- This is the trade-off between **short-term gain and long-term sustainability**; companies, governments and consumers face similar trade-offs.

▸ For whom to produce

- This asks who will use the goods, and how they will be produced and distributed. People differ in needs, income, taste and lifestyle, so producers choose which group to serve.
- Producers study what consumers like, how much they can pay and how much demand exists, so resources are not wasted.

Type of shoe	For whom	Features
School shoes	Students	Simple, durable, affordable
Office-wear shoes	Working professionals	Comfort, formal look, often leather or polished materials
Sports shoes	Athletes and fitness users	Special rubber soles, light materials for grip and support
Casual shoes or slippers	Daily users	Comfortable yet affordable

The target group also affects materials: leather is aimed at office-goers and higher-income buyers, rubber or synthetic at sportspersons, factory workers and people who need affordable, durable footwear.

▸ How to produce

- After deciding what to produce, producers must choose methods, resources and technology, and the right mix of land, labour, capital and technology.

- **Labour-intensive** production uses more workers and less machinery (agriculture, handicrafts). **Capital-intensive** production uses more machines and technology and fewer workers (steel, automobiles).

Factor that influences the choice	How it matters (garment example)
Cost of capital	If machines are costly, the firm uses more labour; if they become cheap, it may automate
Level of technology	Advanced technology encourages machines; limited technology means manual work
Nature of the product	Designer or customised clothes need skilled labour; mass-produced garments suit machines
Cost and availability of labour	Cheap, plentiful labour favours labour-intensive methods; costly or scarce labour favours machines
Laws and regulations	Labour laws or incentives for machinery shape the decision

Economic Systems and How Choices Are Made

The answers to the three questions depend on how resources are organised and **who controls the decisions**. The **economic system** is the arrangement that defines how goods, services and resources are produced, consumed and distributed in a country. There are three kinds.

Feature	Planned economy	Market economy	Mixed economy
Who decides	A **central planning authority** of the government (for example a planning commission)	**Demand and supply** forces with little government interference	Private individuals, enterprises **and** the government together
Ownership	Government owns most land, factories, banks and transport	Mostly **private** individuals and companies own resources	Private ownership plus a large public sector
Prices and output	Fixed by government; firms follow central targets, not demand	Prices and output set by market forces; government does not control them	Private players are regulated by government; some goods are state-supplied
Competition	Strict permits and licences restrict private firms; little competition	Many producers offer similar goods, so quality, lower prices and innovation are encouraged	Private and public firms coexist and compete
Weakness or role of government	Little motive to improve quality or innovate	Government acts like a referee: law and order, safety, **public goods** and physical infrastructure	Fair competition rules, consumer protection, transparency, public goods, welfare programmes
Examples	Former Soviet Union, North Korea, Cuba	USA, Japan, Hong Kong	India (after 1991), China (after 1978), Germany, Sweden

DEFINITION

Public goods are goods and services available to all without anyone being excluded, and one person's use does not prevent others from using them — for example parks, roads, police services, street lights and basic education.

Pure systems rarely exist. Even market economies such as the USA and Singapore have major government involvement, so **almost all economies are mixed**. The government's role in the market can be compared to a referee in a football match.

▸ India's economic journey

- After Independence India followed a **state-led approach**, close to a planned economy: the government controlled industries, allocated resources and regulated production through licences and permits, and banking, transport and heavy industries were dominated by the public sector.
- By **1991** the country faced serious economic difficulties, so major reforms were introduced: less regulation, encouragement to private enterprise, opening to global trade and investment, and more competition.
- India therefore moved towards a more market-oriented system while keeping an important role for the government (a mixed economy).

Quick Revision: Likely Exam Questions

Likely question	Model answer
Differentiate between needs and wants.	Needs are essentials such as food, water and shelter; wants are things we desire, such as gadgets or holidays. Wants are unlimited and keep changing.
What is opportunity cost? Give an example.	The value of the next best alternative given up when a choice is made. Growing more barley means giving up wheat, as in the farmer's table.
What does a PPC show?	The different combinations of two goods that can be produced with all available resources used efficiently; its downward slope shows the trade-off.
Why does scarcity arise?	Because wants are unlimited while resources such as land, labour, capital and money are limited and have alternative uses.
What is the meaning of the word 'economics'?	From the Greek *oikonomia*: *oikos* (household) and *nemein* (management), i.e. household management.
List the three key questions of economics.	What to produce, how to produce, and for whom to produce.
Distinguish between labour-intensive and capital-intensive production.	Labour-intensive uses more workers and less machinery (handicrafts, agriculture); capital-intensive uses more machines and fewer workers (steel, cars).
Which factors decide the choice of production technique?	Cost of capital and labour, level of technology, nature of product, and laws and regulations.
Compare planned and market economies.	Planned: government decides, owns resources, limited competition, little innovation. Market: demand and supply decide, private ownership, competition leads to quality and innovation.

Likely question	Model answer
Why are most economies called mixed economies?	They combine private enterprise and market forces with government regulation, public goods and welfare; pure systems rarely exist.
What are public goods?	Goods and services available to everyone without exclusion, such as roads, parks and street lights.
Why was 1991 important for India?	A serious economic crisis led to reforms that reduced regulation, encouraged private firms, opened trade and increased competition.

★ KEY TAKEAWAYS

- Economics is about **choices** made with **limited resources** to satisfy **unlimited wants**.
- Every choice has an **opportunity cost**; the **PPC** shows the trade-off between two goods.
- Three key questions: **what, how and for whom** to produce.
- **Planned** economies rely on government control, **market** economies on demand and supply, **mixed** economies on both.
- Almost every economy today, including **India after 1991**, is a mixed economy.
- Good economic decisions rest on **data and analysis**.

CHAPTER 02 · ECONOMICS

The Price Puzzle: What Drives the Market

Mango prices fall mid-season, hotel rooms cost more at New Year, and masks became costly in 2020. Prices are not random: they respond to two forces, demand and supply, and sometimes to the government. This chapter explains how.

Big Questions

- What factors influence the demand for and supply of goods and services in a market?
- How are prices determined through the interaction of demand and supply?
- What is market equilibrium, and does it exist in the real world?
- How and why does the government intervene in the market?

Why Do Prices Change?

Prices react to what people want, how much is available, seasons, festivals, trends and even rumours. For every good or service, from snacks to mobile phones, price is decided by **demand** and **supply** acting together. In Grade 7 you saw that sellers adjust prices that are too high or too low; this chapter looks at the forces behind that.

Demand

DEFINITION

Demand is the quantity of a product that people are **willing and able** to buy at a particular price. Desire alone is not demand: it must come with **purchasing power** (how much one unit of a currency can buy).

▸ Law of Demand

- When the price of a good rises, quantity demanded **falls**; when price falls, quantity demanded **rises**.
- This **inverse relationship** between price and quantity demanded is the Law of Demand (mangoes bought in smaller quantities at the start of the season, in larger quantities as prices fall).

▸ Individual demand: the demand schedule and curve

Individual demand is the quantity one consumer wants to buy at different prices, with other factors constant. The table of prices and quantities is the **demand schedule**; drawing it on a graph gives the **demand curve**.

Price of mangoes (₹/kg)	Quantity demanded by Srivalli	Point on graph
150	1 kg	A
100	2 kg	B
50	3 kg	C

- Price goes on the y-axis and quantity on the x-axis. Joining A, B, C gives a **downward-sloping line DD'**, the demand curve, which shows the inverse price–quantity relation, assuming income, taste etc. are constant.

▸ Market demand

Market demand is the total quantity demanded by all potential buyers at each price, i.e. the **sum of all individual demands**.

Price	Srivalli (Q1)	Alex (Q2)	Israt (Q3)	Market demand (QD = Q1+Q2+Q3)
₹150	1 kg	2 kg	3 kg	6 kg
₹100	2 kg	4 kg	6 kg	12 kg
₹50	3 kg	6 kg	9 kg	18 kg

REMEMBER

The market demand curve is **flatter** than an individual curve because many buyers respond to the same price change together. When price falls from ₹150 to ₹50, Srivalli buys 2 kg more, but the market buys 12 kg more.

▸ Other determinants of demand

Demand can change even when price does not, as with the rush for a new smartphone. The main factors are:

Determinant	How it changes demand	Example
Price of related goods: substitutes	Goods that replace each other: if one becomes dearer, demand for the other rises	Tea and coffee; mangoes and bananas
Price of related goods: complements	Goods used together move together: demand for one rises when the other's demand rises	Smartphones and earphones; cars and petrol; printers and cartridges; cinema tickets and popcorn
Income of the consumer	Higher income means people buy more or pick better quality even at the same price	Moving to branded products
Taste and preference	Personal liking decides demand	Srivalli likes mangoes and will not switch to cheaper oranges
Population size and composition	More people raise demand; age mix shapes the type of goods	More children: sports shoes; more working adults: formal shoes; more elderly: orthopaedic shoes. India's large population drives domestic demand
Seasonality	Weather, festivals and culture shift demand through the year	Books at the start of the session, sweets at festivals, sweaters in winter
Future price expectations	If prices are expected to fall, buyers wait; if to rise, they buy now	Postponing durable purchases before Diwali or New Year for discounts

DEFINITION

Diminishing marginal utility: the extra satisfaction from each additional unit of a good keeps falling (the first mango is delicious, the fifth hardly interesting). Lower utility means lower willingness to pay, so demand falls.

Supply

DEFINITION

Supply is the quantity of a product that sellers are **willing and able** to offer at a particular price.

▸ Law of Supply

- As price rises, quantity supplied **rises**; as price falls, it **falls** (a **direct relationship**).
- Reason: higher prices mean higher profit, which encourages producers to produce more and attracts new firms.

At the start of the mango season supply is low and mangoes are costly; mid-season supply rises and price falls. **When supply is less than demand, prices rise; when supply exceeds demand, prices fall.**

▸ Individual and market supply

Price	Seller A	Seller B	Seller C	Market supply (A+B+C = QS)
₹50	1 kg	3 kg	2 kg	6 kg
₹100	2 kg	4 kg	6 kg	12 kg
₹150	3 kg	7 kg	8 kg	18 kg

- Individual supply is what one seller offers at different prices; **market supply** is the sum of all sellers' supplies.
- Plotting price against quantity gives an **upward-sloping supply curve** (SS' for one seller, SMSM' for the market).

▸ Other determinants of supply

Determinant	Effect	Example
Price of related goods	Producers shift to the more profitable alternative	Wheat price low and chickpea price high: the farmer plants more chickpeas next season
Number of sellers	More sellers mean more competition and higher market supply, pulling prices down; fewer sellers do the opposite	A market flooded with sellers
Technology	Better technology lowers the cost of production and raises supply (and the reverse)	Drip irrigation and weather sensors raise crop output; cold storage helps transport
Future expectations	Expected boom in demand raises output; expected fall lowers it; sellers may hold back stock	Potato wholesalers holding stock to sell at higher peak-season prices
Other factors	Cost of inputs, discovery of alternative inputs, depletion of resources, weather and disasters change supply	Drought reduces crop supply

Market Equilibrium

A market is a negotiation between what buyers are willing to pay and what sellers are willing to accept, so prices come from the interaction of demand and supply.

Price (₹)	Quantity demanded (kg)	Quantity supplied (kg)	Outcome
40	38	6	Qs < Qd: **excess demand** (shortage)
100	12	12	Qs = Qd: **market equilibrium**
150	8	43	Qs > Qd: **excess supply** (surplus)

- At ₹100 quantity demanded equals quantity supplied. This is **market equilibrium**: equilibrium price = ₹100, equilibrium quantity = 12 kg. The market is 'cleared': no shortage, no surplus, no pressure on price.
- On the graph, equilibrium is the point **E** where the demand curve DMDM' intersects the supply curve SMSM'.
- At a price below equilibrium there is excess demand and price tends to rise; above equilibrium there is excess supply and price tends to fall, so the market moves toward equilibrium.

DEFINITION

Market equilibrium is the point where demand equals supply, with neither excess supply (surplus) nor excess demand (shortage); prices stay stable unless external factors change.

▸ Does equilibrium exist in the real world?

- In theory equilibrium is a fixed intersection point, but real markets are **dynamic**: technology, wages, interest rates, wars, political events, pandemics, weather and disasters keep shifting demand and supply.
- So equilibrium is never stable; the market is always adjusting to a **new** equilibrium.
- **COVID-19 example (2020):** demand for face masks surged, supply could not catch up quickly and prices rose; later suppliers adjusted and prices fell; after the pandemic demand dropped and prices returned to earlier levels.

▸ Hotel tariffs: a dynamic market

Situation (hotel with 100 rooms in Goa)	Tariff per night
Off-season weekday (Monday in July)	₹1,500
Weekend in tourist season (Saturday in December)	₹8,000
New Year's Eve (very high demand)	₹25,000
Group tour cancels	Tariff may be cut by 40 per cent overnight to fill rooms

Hotels may change tariff several times a day to maximise **revenue** (total money earned from sales before expenses). They watch booking speed, rival hotels' tariffs, festivals or conferences in the area, weather forecasts, days left before arrival and past booking trends.

NOTE

Choices made today affect future supply: fast-fashion demand, overfishing and overuse of groundwater can reduce future supply. Short-term gain must be balanced with long-term sustainability, which also shapes equilibrium.

Role of Government in the Economy

India is a market-based but **regulated** economy, and the fourth-largest economy in the world. Markets allocate goods by willingness and ability to pay, which may leave poor people unable to afford essentials such as medicines, so fairness and equity need the government's attention.

▸ 1. Regulation of unfair practices

- The government protects consumers, workers and producers from exploitation. A **price ceiling** is a maximum price (for example for medicines) to stop overcharging. A **price floor** is a minimum price (for example a minimum wage); to be effective a price floor must be set **above** the equilibrium price.
- **Monopoly** (one seller, or a few, controlling supply with no close substitutes) allows higher prices, poorer quality and restricted supply; the government keeps price and quantity in check.
- Regulators ensure transparency: **RBI** (banking), **Central Consumer Protection Authority** (consumer rights and unfair trade), **TRAI** (telecom), **SEBI** (securities market).

> **IMPORTANT**
>
> **Sanitiser case (COVID-19):** demand surged, stocks ran out and prices soared; some shopkeepers resorted to **hoarding** and **black marketing**. The government declared sanitisers essential commodities under the **Essential Commodities Act, 1955** and capped the price at ₹100 for a 200 ml bottle. Production then rose and supply reached fair prices. Such controls suit emergencies, not forever.

Term	Meaning
Hoarding	Accumulating goods beyond immediate need, driven by fear of shortage or expected price rise
Black marketing	Illegal trade in goods and services that are banned or regulated
Price ceiling	Maximum price a seller can charge
Price floor	Minimum price that can be charged
Monopoly	Single seller controlling the whole supply of a product with no close substitute

▸ 2. Provision of public goods

- **Public goods** (roads, bridges, parks, street lights, national defence, sanitation and drainage) benefit all citizens and are usually not offered by private firms because they bring no direct profit.
- **Free-rider problem:** if each family is asked to contribute ₹5,000 to build a park, many think others will pay and they can still use it, so too little money is collected and the park is never built. Hence the government must provide or fund such goods for welfare, development and equal access.

▸ Limitations of government intervention

Problem	Explanation	Example
Price distortions and reduced producer incentives	Prices fixed below market level reduce the motive to supply, causing shortages	Maximum wheat price of ₹20/kg when the market price is ₹30/kg
Compliance burden	Too many licences, permits and rules hurt small firms and the **ease of doing business**	A small restaurant needing food, fire, pollution and local clearances
Discourages innovation and entrepreneurship	Price controls and heavy regulation reduce the return on investing in new ideas	Farmers not investing in better seeds or irrigation

> **NOTE**
>
> **Ease of doing business** measures how simple it is to start, run and close a business, depending on bureaucratic control, efficiency and legal frameworks. A democratic government is accountable to the people and must balance the voices of consumers, producers and workers while intervening.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is demand? How does it differ from desire?	Demand is the quantity people are willing and able to buy at a price. Desire without purchasing power is not demand.
State the Law of Demand.	Other things constant, quantity demanded falls when price rises and rises when price falls: an inverse relationship.
Why is the demand curve downward sloping?	Higher price lowers quantity demanded and lower price raises it; it shows the inverse relation.
Distinguish individual demand from market demand.	Individual demand is one buyer's quantity at various prices; market demand is the sum of all buyers' demand.
Name four determinants of demand other than price.	Price of related goods, income, taste and preference, seasonality, population, future price expectations.
Distinguish substitutes from complements.	Substitutes replace each other (tea, coffee); complements are used together (smartphone, earphones).
State the Law of Supply and give reasons.	Quantity supplied rises with price because higher prices bring more profit and attract new firms.
List factors that determine supply.	Price of related goods, number of sellers, technology, future expectations, input costs, weather.
What is market equilibrium?	The price and quantity at which demand equals supply, with no shortage or surplus. In the table: ₹100 and 12 kg.
What happens at a price above equilibrium?	Excess supply (surplus), so price tends to fall toward equilibrium.
Why is equilibrium never stable in the real world?	Technology, weather, wars, pandemics and other events keep changing demand and supply (masks in 2020, hotel tariffs).
Why does the government intervene in markets?	To prevent exploitation, control monopolies and hoarding, ensure essentials are affordable and provide public goods.
Differentiate a price ceiling from a price floor.	Ceiling: maximum price (medicines). Floor: minimum price (minimum wage), effective only above equilibrium.
What are the drawbacks of excessive regulation?	Distorted prices and shortages, compliance costs for small firms, and weaker innovation.

★ KEY TAKEAWAYS

- **Demand** and **supply** together determine price.
- **Law of Demand:** price and quantity demanded move in **opposite** directions. **Law of Supply:** they move in the **same** direction.
- Market demand and market supply are **sums** of individual demand and supply.
- Demand is shifted by related goods, income, taste, population, seasons and expectations; supply by related goods, number of sellers, technology and expectations.
- **Equilibrium** is where demand equals supply (₹100, 12 kg in the mango example), but real markets keep moving.
- Governments regulate unfair practices, check monopolies, set price ceilings and floors and provide **public goods**; excessive control causes distortions.

CHAPTER 03 · GEOGRAPHY

Shaping the Earth's Surface

The Earth's surface is never still. Forces from deep inside the planet push up mountains and shake the ground, while forces on the surface wear them down and rebuild them elsewhere. This chapter explains the theory of plate tectonics, the work of weathering and erosion, the landforms made by water, waves, ice, wind and groundwater, and the disasters linked to them.

Big Questions

- What shapes the Earth's surface?
- What is plate tectonics, and what are the effects of plate movement?
- How are landforms formed and classified?
- How are humans and other living beings connected to landforms?
- How do disasters linked to different landforms affect human lives?

Landforms and Plate Tectonics

DEFINITION

A **landform** is a natural feature of the Earth's surface made by processes such as weathering, erosion, deposition and movements of the Earth. Examples: mountains, valleys, plateaus, plains, deserts, coastal features.

Plate tectonics, a theory associated with **W.J. Morgan**, explains the movement of the Earth's crust. The outermost layer is not one piece but is broken into large and small **tectonic plates**, which move slowly over the semi-molten layer below. They are responsible for mountains, earthquakes and volcanoes, and explain the distribution of continents and oceans.

▸ Interior of the Earth

Layer	Features
Crust	Outermost solid layer on which we live. Thickness: 30–40 km under continents (continental crust, about 30 km in the figure), 5–7 km under oceans (oceanic crust)
Lithosphere	Rigid outer layer: the crust plus the upper part of the mantle (about 100 km). It is broken into tectonic plates
Asthenosphere	Hot, mobile, partially molten layer (about 200 km) beneath the lithosphere that allows plates to move
Mantle	Very thick, hot, mostly solid layer between crust and core (about 2,900 km)
Outer core	Fluid layer mainly of iron and nickel (about 2,200 km)
Inner core	Solid, hot, spinning metal ball; the densest part of the Earth (about 1,250 km)

NOTE

All values are approximate. Move down the layers: crust → mantle → outer core → inner core.

▸ Plates and why they move

- Tectonic plates are huge slabs of solid rock that move only **a few centimetres a year**.
- Types: **continental** plates (carry continents), **oceanic** plates (carry ocean floors) and **mixed** plates (carry both).
- Major plates: Pacific, Eurasian, African, North American, South American, Indo-Australian and Antarctic; there are also minor plates.
- **Cause:** heat from the core makes hot material in the mantle rise while cooler material sinks. These **convection currents** push and pull the plates in different directions.

▸ Plate boundaries

Boundary	Plate movement	Result	Example
Convergent	Plates move towards each other	Continent–continent collision forms **fold mountains**; oceanic plate sinks beneath a continental plate, causing volcanoes and earthquakes	Himalaya
Divergent	Plates move apart	Magma rises and forms new crust, making **mid-ocean ridges**	Mid-Atlantic Ridge
Transform	Plates slide past each other	No crust created or destroyed; mainly earthquakes	San Andreas Fault, USA

IMPORTANT

Most earthquakes and volcanoes occur along plate boundaries, especially around the Pacific Ocean, an area called the **Ring of Fire**. Comparing the plate map with the earthquake and volcano map shows this link. The theory helps identify earthquake and volcano-prone regions.

▸ Earthquakes and volcanoes: India and the past

- India has had major earthquakes with thousands of deaths, such as the **Gujarat earthquake of 2001**. Dense population makes damage severe.
- Earthquakes were known in early India as **bhūkampa** ('shaking of the Earth'). **Varāhamihira**, in the **Bṛihatsaṁhitā**, noted that changes in wind, rain, clouds, animal behaviour and planetary alignments could signal them, and linked them to four elemental forces: Vāyu (wind), Agni (fire), Indra (heaven/thunder) and Varuṇa (water). It was an early blend of observation and cosmological reasoning.
- Volcanic eruptions deposit grey ash, showing the Earth's internal forces at work.
- **Baratang Island** (Andaman and Nicobar Islands) has India's only **mud volcano**: mud bubbles out due to underground gases and pressure instead of fiery lava.

Weathering and Erosion

Together, weathering and erosion wear down mountains, carve valleys, form plains and create caves, cliffs and deltas.

	Weathering	Erosion
Meaning	Breaking down of rocks into smaller pieces at the surface	Wearing away and **carrying** of soil and rock from one place to another
Movement of material	No movement	Involves movement by agents: water, wind, ice, waves

	Weathering	Erosion
Types	Physical, chemical, biological	Water, wind, glacial, coastal

Type of weathering	Cause	Example
Physical	Temperature change (heating and cooling expansion), frost, wind	Rocks crack into smaller pieces
Chemical	Minerals react with water, air or acids to form new substances	Rain and air altering rock minerals
Biological	Plants, animals, micro-organisms	Plant roots growing into cracks and splitting rock

Weathering also forms **soil**. Types of erosion: **water** (rivers, rain, waves), **wind** (dry, sandy areas), **glacial** (moving ice scrapes and carries rocks) and **coastal** (sea waves wear the shore).

▸ Erosion and human livelihoods

- **Farmers:** fertile topsoil is removed, so yields fall.
- **River and coastal dwellers:** land, houses and roads are washed away.
- **Construction and mining:** unstable land creates safety risks.
- **Tourism and fishing:** beaches, rivers and fertile lands are destroyed.

Soil conservation method	Meaning
Contouring (continuous contour trench)	Trenches dug along contour lines of a hillside to slow water, let rain soak in, prevent erosion and recharge groundwater
Bunding	Earthen embankments along contour lines to slow run-off, reduce erosion and increase infiltration and soil moisture
Terracing	Level or gently sloping steps cut on a hillside to prevent soil erosion

> **NOTE**
> The **Sindhu-Sarasvatī civilisation** used contouring, bunding, terracing, dams and canals. Texts such as the Vedas, Kṛiṣhiparāśhara, Kauṭilya's Arthaśhāstra (land assessment by fertility) and Vṛikṣhāyurveda record such practices. The **Zabo system** of Nagaland uses earthen bunds on hillslopes for soil and water conservation; **check dams** across small streams slow water, prevent erosion, let sediment settle and recharge groundwater.

Agents of Gradation

Agents of gradation wear down, transport and deposit material, levelling the surface over time: **running water, glaciers, wind, waves and groundwater**. They lower high areas and fill low areas.

Agent	Main work
Running water	Erodes rocks and soil, forming valleys and plains
Glaciers	Scrape and carry huge amounts of material; carve U-shaped valleys
Wind	Shapes deserts by eroding and depositing sand

Agent	Main work
Sea waves	Erode coasts to form cliffs, beaches, bays
Groundwater	Dissolves limestone, creating caves and sinkholes

> **NOTE**
> Landforms have shaped history. Rivers and plains (Ganga, Nile, Brahmaputra, Indus) gave rise to farming societies and cities. The Himalayas shielded India yet allowed exchange through passes like the **Khyber Pass**. Deserts like the **Thar** limited large settlements but encouraged trade routes like the **Silk Route**. Coasts and harbours supported trade and kingdoms in south India.

▸ Running water

Course of river	Process	Landforms
Upper	Steep gradient, strong erosion	V-shaped valleys, waterfalls, rapids
Middle	Loses energy, begins to deposit	Meanders, oxbow lakes, floodplains
Lower	Slows further, heavy deposition	Deltas, levees, alluvial fans

Landform	Formation	Human importance
Waterfall	River falls over a steep drop in the upper course where hard rock resists erosion and softer rock below is worn away	Tourism, hydroelectric power, trekking and photography, religious and cultural value
Meander	Winding bend of middle/lower course: river erodes the outer bank and deposits sediment on the inner bank; a cut-off loop becomes an **oxbow lake**	Fertile soil for farming, settlements on gentle slopes, navigation, irrigation, tourism. Example: Grand Anicut (Kallanai), Tamil Nadu, irrigation
Delta	At the river mouth where it meets sea/lake, sediment builds up into a fan-shaped or triangular area, with distributaries	Very fertile (rice, jute), fishing, dense settlement, trade routes; but flood-prone. Example: Sundarbans delta

▸ Waves and currents (coasts)

- Waves and currents create beaches, sand bars, sea cliffs, caves, arches and stacks.
- A **beach** is a deposit of sand, pebbles or rock along a shore. It supports tourism, fishing, shell and sand collection and acts as a natural barrier against waves and erosion.

Coastal erosion landform	Meaning
Cliff	Steep rock face formed as waves undercut the base
Wave-cut (shore) platform	Flat area left as cliffs retreat
Cave	Formed as waves erode weak parts of rock
Arch	Caves on opposite sides of a headland meet
Stack	Isolated rock pillar left when the arch collapses

▸ Glaciers

Landform	Meaning
U-shaped valley	River valley widened and deepened by a glacier
Cirque	Bowl-shaped hollow at the head of a glacier
Arête	Sharp ridge between valleys
Hanging valley	Where a smaller glacier meets a larger one
Fjord	Deep, narrow sea inlet formed when the sea floods a glacial valley

Human use: U-shaped valleys and cirques for trekking, skiing, mountaineering; fjords for harbours and fishing; fertile glacial soil; glaciers as sources of fresh water feeding rivers.

Moraine (deposited debris called till)	Where formed
Lateral	Along the sides of a glacier
Terminal	At the end, marking the furthest advance
Medial	In the middle, when two glaciers meet and their lateral moraines join

Moraines give fertile soil and can form natural dams and lakes for water supply, irrigation and hydroelectric power. Case: the **Chamoli flood, Uttarakhand, February 2021**, killed people and livestock and damaged buildings, roads, bridges and hydel projects.

▸ Wind

Landform	Meaning
Yardang	Streamlined rock ridge carved by wind
Ventifact	Rock polished and shaped by sandblasting
Deflation hollow (blowout)	Shallow depression where loose material is blown away (may expose water table: oasis)
Desert pavement	Flat surface left after fine particles are blown away

Dune type	Features
Barchan	Crescent-shaped; limited sand, single wind direction
Longitudinal	Long ridges parallel to prevailing wind
Star	Many arms; winds from different directions
Parabolic	U-shaped, often stabilised by vegetation

Dunes act as barriers against desertification and wind erosion, protect coastal settlements from sea winds, give adventure sports and supply construction sand.

▸ Underground water (karst topography)

In limestone and other soluble rocks, acidic water dissolves rock by chemical weathering and erosion, giving **karst topography**.

Landform	Meaning
Cave	Hollow space dissolved by acidic water
Stalactite	Icicle-shaped formation hanging from the cave ceiling
Stalagmite	Formation rising from the cave floor
Sinkhole (doline)	Depression where ground collapses into an underground cavity
Underground river	Flows through cave systems

Caves and underground rivers provide fresh water, tourism, cultural or religious significance; stalactites and stalagmites attract geologists and adventurers.

Landforms and Disasters

Disaster	Causes	Human factors / notes
Landslide	Heavy continuous rain (adds weight, reduces friction); earthquakes and eruptions; steep slopes; loose or weathered rock	Deforestation, mining, road building, unplanned hillside construction, poor drainage and improper land use
Avalanche	Heavy snowfall in a short time; weak snow layers; sudden temperature rise (melting reduces friction); strong winds piling snow; earthquakes and vibrations	Skiing, trekking, construction in mountains
GLOF (Glacial Lake Outburst Flood)	Rapid glacier melt from rising temperatures strains ice/moraine dams; heavy rain or snow; earthquakes, avalanches or landslides weaken the dam, releasing water suddenly	Destructive floods downstream (compare Chamoli 2021)
Dust storm	Strong winds lift loose dry soil; prolonged drought; desert and semi-arid regions; sparse vegetation	Deforestation, overgrazing, poor farming; climate change raises frequency

REMEMBER

For each disaster, be ready to list prone areas, mitigation measures and recent examples from India and the world. Mitigation includes afforestation, better drainage and land-use planning, early warning, monitoring glacial lakes, and awareness and preparedness drills.

NOTE

Internal (endogenic) forces: earthquakes, volcanoes, folding and faulting build mountains, valleys and ocean basins. **External (exogenic) forces**: weathering, erosion and deposition wear them down and reshape them.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is plate tectonics?	A theory (W.J. Morgan) that the crust is broken into plates that move slowly over the semi-molten asthenosphere, forming mountains, volcanoes and earthquakes.

Likely question	Model answer
Describe the layers of the Earth.	Crust, mantle (with the asthenosphere below the lithosphere), outer core (liquid iron-nickel), inner core (solid, densest).
What causes plate movement?	Convection currents in the mantle driven by heat from the core.
Name and explain the three plate boundaries.	Convergent (move together; Himalaya), divergent (move apart; Mid-Atlantic Ridge), transform (slide past; San Andreas Fault).
What is the Ring of Fire?	Zone around the Pacific Ocean with most earthquakes and volcanoes, along plate boundaries.
Differentiate weathering from erosion.	Weathering breaks rock in place; erosion removes and transports material.
Name the types of weathering.	Physical, chemical and biological.
How does erosion affect livelihoods? Name soil conservation methods.	Farmers lose topsoil, coasts lose land, etc. Methods: contouring, bunding, terracing, check dams.
List the agents of gradation.	Running water, glaciers, wind, waves, groundwater.
Explain how a delta and a meander form, and why they matter.	Delta: sediment deposition at river mouth; meander: erosion on outer bank, deposition on inner. Both give fertile land but delta is flood-prone.
Name landforms made by glaciers and by wind.	Glacial: U-valley, cirque, arête, hanging valley, fjord, moraines. Wind: yardang, ventifact, deflation hollow, desert pavement, dunes.
What are karst landforms?	Caves, stalactites, stalagmites, sinkholes and underground rivers formed by dissolving of limestone.
Explain GLOFs.	Sudden floods when glacial lake dams of ice or moraine fail due to melting, heavy rain or shocks.
How do human activities cause landslides?	Deforestation, mining, road building, unplanned construction and poor drainage.

★ KEY TAKEAWAYS

- The Earth has crust, mantle and core; the lithosphere is broken into tectonic plates that move by mantle convection.
- Plate boundaries (convergent, divergent, transform) explain mountains, ridges, earthquakes and volcanoes, concentrated in the Ring of Fire.
- Weathering breaks rock; erosion carries it away; deposition builds new landforms.
- Running water, waves, glaciers, wind and groundwater are the agents of gradation, each with characteristic landforms.
- Landforms shape settlement, farming, trade, tourism and history, while landslides, avalanches, GLOFs and dust storms threaten life.
- Soil conservation (contouring, bunding, terracing) and preparedness reduce damage.

CHAPTER 04 · GEOGRAPHY

Atmosphere and Climate

Clouds, sunshine and breeze are all products of the blanket of air around the Earth. This chapter covers what the atmosphere is made of, how it is layered, how weather differs from climate, the seasons and monsoon of India, climate change, and the 2025 Punjab floods as a case study.

Big Questions

- What is the composition of the atmosphere?
- How do the layers of the atmosphere affect the Earth?
- What is the mechanism of the monsoon?
- How can we reduce our carbon footprint?

The Atmosphere

The **atmosphere** is the blanket of air held around the Earth by **gravity** (the force of attraction the Earth exerts on objects on or near its surface). It is a mixture of gases vital for life. It:

- shields us from harmful radiation of the Sun, including ultraviolet rays;
- regulates the Earth's temperature by trapping some of the Sun's energy and preventing it from escaping to space;
- is a key part of weather and climate, influencing temperature, humidity and air pressure.

Composition and Structure of the Atmosphere

▸ Composition

Gas	Share of atmosphere
Nitrogen	78%
Oxygen	21%
Argon	0.93%
Carbon dioxide	0.04%
Others (helium, neon, krypton, xenon, ozone, hydrogen)	0.03%

- Besides gases there are **water vapour** (generally 0.1–0.4%; essential for clouds and precipitation) and tiny **dust particles**.
- Composition varies with altitude. Nitrogen is useful to plants (recall Grade 8 Science).

▸ Structure: five layers

Layers are defined by changes in temperature and density with height. **Altitude** is the height of a place above mean sea level (taken as zero), measured in metres or feet.

Layer	Height	Temperature	Key features
Troposphere	Average about 12 km	Falls with altitude	Most important layer: air we breathe, most water vapour and clouds, **all weather** (rain, fog, hail). Top boundary: **tropopause**
Stratosphere	Up to about 50 km	Rises (ozone absorbs UV)	Cloud-free and calm, ideal for **aeroplanes**; contains the **ozone layer** that filters harmful UV radiation. Boundary: **stratopause**
Mesosphere	Up to about 80 km	Falls with altitude	Most **meteors** from space burn up here. Boundary: mesopause
Thermosphere	About 80–700 km	Rises very rapidly (absorbs X-rays and short-wave UV)	Reflects **radio waves** back to Earth; contains the **ionosphere**; **auroras** occur here
Exosphere	Uppermost	—	Very thin air; light gases (helium, hydrogen) escape into space because of weak gravity

REMEMBER

Temperature decreases with altitude only in the troposphere and the mesosphere.

DEFINITION

Aurora: colourful light display near the poles. Charged particles of the **solar wind** are guided to the magnetic poles and make atmospheric gases glow. Called **Aurora Borealis** in the Northern Hemisphere and **Aurora Australis** in the Southern. 'Aurora' is Latin for dawn, after the Roman goddess of dawn.

NOTE

Why we do not feel air pressure: air presses on us from all sides and the pressure inside our bodies equals atmospheric pressure, so the two cancel.

Weather and Climate

	Weather	Climate
Meaning	Hour-to-hour and day-to-day condition of the atmosphere	Sum total of weather conditions and variations over a large area for a long time, usually **30 years or more**
Variation	Changes quickly, even within a day	Fairly stable average pattern

▸ Elements of weather and climate

Element	Key points
Temperature	Varies by day/night and season. Governed by **insolation** (incoming solar energy intercepted by the Earth), which decreases from the equator to the poles, so temperature also falls from equator to poles (torrid, temperate and frigid zones)
Humidity	Water vapour in the air from evaporation. Warm air holds more vapour; high humidity means clothes dry slowly and sweat evaporates slowly, so we feel uncomfortable

Element	Key points
Precipitation	Air saturated with water vapour condenses and falls: drizzle, rain, snow, sleet, hail. Affected by prevailing winds, mountains and seasons. Rain lowers temperature and recharges groundwater; long absence gives a dry climate
Atmospheric pressure	Weight of air on the surface; highest at sea level, falls rapidly with altitude. Warm air rises → **low pressure** (cloudy, wet); cold air sinks → **high pressure** (clear, sunny)
Wind	Air moving from high to low pressure; direction influenced by the Earth's rotation. Winds are named by the direction they blow **from** (westerly = from the west)

Wind category	Speed (km/hr)	Effect
Calm	0–1	Smoke rises vertically
Light breeze	6–11	Leaves rustle; vane moves
Strong breeze	39–49	Large branches sway; umbrellas hard to use
Storm	103–117	Widespread damage

▸ Land and sea breeze (local winds)

	Sea breeze	Land breeze
When	Day, especially afternoon	Night
Cause	Land warmer than sea: low pressure over land	Land cools faster than sea; sea relatively warmer
Direction	Sea → land	Land → sea
Strength	Stronger (large pressure difference)	Weaker (small difference)

These moderate the climate of coastal regions.

Seasons in India

India's climate is **tropical monsoon**. The **Indian Meteorological Department (IMD)** recognises four seasons:

Season	Period	Features
Winter	December to early April	Coldest months December–January; north-west about 10–15°C; south-east about 20–25°C
Summer / pre-monsoon	April to June (to July in the north-west)	Hottest month April in west/south, May in north; inland 32–40°C
Monsoon / rainy (advancing)	June to September	Humid south-west summer monsoon sweeps the country from late May–early June; recedes from north India at the start of October
Post-monsoon (retreating)	October to December	North-west mostly cloudless in October–November; south India gets more rain. Himalayan states also have **autumn and spring**

▸ Traditional Indian seasons (Ṛtu)

Traditionally India has **six seasons**, each about two months long, from the astronomical division of 12 months into six parts.

Ṛtu	Indian months	Gregorian months
Vasanta (spring)	Chaitra–Vaiśhākha	March–April
Grīṣhma (summer)	Jyeṣhṭha–Āṣhāḍha	May–June
Varṣhā (monsoon)	Śhrāvaṇa–Bhādrapada	July–August
Śharad (early autumn)	Āśhvina–Kārtika	September–October
Hemanta (late autumn)	Mārgaśhīrṣha–Pauṣha	November–December
Śhiśhira (winter)	Māgha–Phālguna	January–February

> **NOTE**
> Kauṭilya's Arthaśhāstra records scientific rainfall measurements used in managing revenue and relief. Hindustani classical music links particular rāgas with seasons.

Monsoon

Monsoon is the seasonal reversal of wind direction during a year. Sailors on ancient sailing ships noticed it; the **Arabs** named it from the Arabic **mausim** ('season').

	South-west (summer) monsoon	North-east (winter) monsoon
Period	June to September	October to February
Direction	Sea → land over the Indian Ocean, Arabian Sea and Bay of Bengal	Land → sea
Cause	Land heats faster than ocean in summer: **low pressure** over the subcontinent, **high pressure** over the cooler ocean. Moist winds blow inland	Land cools faster than ocean: **high pressure** over land, low over seas
Rainfall	Brings **most of India's rainfall**	Cold, dry winds, generally no rain; but over the Bay of Bengal they pick up moisture and bring rain to the **east coast**: Tamil Nadu, Andhra Pradesh, parts of Karnataka

> **IMPORTANT**
> **Importance of monsoon:** most Indian agriculture depends on it; it governs food production and water in rivers, reservoirs and wells; it affects transport, festivals and rural jobs. Excess rain causes floods; weak monsoon causes drought. Hence it shapes the economy and livelihoods and gives the country climatic unity.

▸ Knowledge traditions and missions

- **Kṛiṣhiparāśhara** predicted rainfall from positions of the Moon and Sun; **Varāhamihira's Bṛihatsaṁhitā** used **nakṣhatras** (lunar mansions: sky divided into 27 parts along the Moon's path; the Moon takes about 27 days to circle and moves through about one nakṣhatra a day). Many Indian farming practices still follow these methods.
- **Kālidāsa's Meghadūtam** (about 5th century CE) mentions the onset date of the monsoon over central India and traces the path of monsoon clouds.
- **National Monsoon Mission (NMM)**, Ministry of Earth Sciences, builds weather and climate prediction models. **Mission Mausam** aims to make India 'Weather Ready' and 'Climate Smart' with better

observation, modelling and forecasting for agriculture, disaster management and rural development.

> **NOTE**
> Monsoon advances: normal onset begins on about 22 May in the far south and reaches north-west India by about 8 July; retreat begins about 17 September from the north-west and ends about 15 October in the south.

Climate Change

Climate change is the long-term change in weather patterns (temperature, rainfall, wind) caused mainly by human activities: burning fossil fuels, deforestation and industrial pollution. These raise **greenhouse gases** (carbon dioxide, methane, nitrous oxide, water vapour), which trap heat and raise global temperatures.

Effects	Details
Environmental	More floods and droughts, melting glaciers, rising sea level, loss of biodiversity
Human	Health, agriculture and livelihoods; affects all sections including women and children

> **DEFINITION**
> **Carbon footprint:** total greenhouse gases released into the atmosphere by human activities such as energy use, transport and production of goods and services.

Ways to reduce it: save energy and switch off appliances, use renewable energy, walk/cycle/use public transport or carpool, avoid single-use plastics and reuse/recycle, save water, protect forests and plant trees, adopt sustainable lifestyles. Self-check categories: transport, electricity, water, waste and plastics (low impact = 1 point to very high = 4 points); then pledge two changes.

Case Study: Punjab Floods, 2025

Heavy monsoon rains and swelling of the rivers **Satluj, Beas and Ravi** (and the Ghaggar) flooded large parts of Punjab, damaging villages, fields, houses, roads and bridges.

Natural causes	Human-made causes
Very heavy monsoon rain, intensified by western disturbances; heavy rain also in Himachal Pradesh and Jammu & Kashmir; rivers already flowing high	Weak, old embankments (**dhūsī bāndh**); houses and farms built too close to rivers; silt in rivers and dams reducing capacity; late or unclear flood warnings

- **Effects:** deaths; thousands moved to relief camps; paddy and other crops damaged; poultry and dairy losses (cows, buffaloes, chickens); roads, bridges, border fences and public buildings damaged; standing water caused waterborne disease and sanitation problems.
- **Lessons:** better flood management, embankment repair, desilting, regulated building near rivers, timely warnings and disaster preparedness (NDMA guidelines).

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is the atmosphere? Give its composition.	Envelope of gases held by gravity: nitrogen 78%, oxygen 21%, argon 0.93%, CO2 0.04%, others 0.03%, plus water vapour and dust.

Likely question	Model answer
Name the layers of the atmosphere in order.	Troposphere, stratosphere, mesosphere, thermosphere, exosphere.
In which layer do aeroplanes fly and why?	Stratosphere: free of clouds and weather disturbances.
Why is the troposphere the most important layer?	It has the air we breathe, water vapour, clouds and all weather phenomena.
Why do we not feel the pressure of the atmosphere?	Air presses from all sides and the pressure inside our body equals it, cancelling.
Distinguish weather from climate.	Weather: day-to-day state of the atmosphere. Climate: average over 30 years or more.
Name the elements of weather and climate.	Temperature, humidity, precipitation, atmospheric pressure, wind.
Explain land and sea breeze.	Day: land warmer, sea breeze to land. Night: land cools faster, land breeze to sea.
What are the four seasons of India (IMD)?	Winter, summer (pre-monsoon), monsoon, post-monsoon.
Distinguish the south-west monsoon from the north-east monsoon.	SW: June–September, sea to land, most rain. NE: October–February, land to sea, dry but rain on the east coast after crossing the Bay of Bengal.
Why is the monsoon important to India?	Agriculture, water supply, economy and daily life; floods or drought if it goes wrong.
Where does the word 'monsoon' come from?	Arabic mausim, meaning season.
What is climate change and what causes it?	Long-term change in weather patterns caused mainly by fossil fuels, deforestation and industrial pollution raising greenhouse gases.
How can we reduce our carbon footprint?	Save energy, use renewables, plant trees, use public transport, reduce plastic and water waste.
What caused the 2025 Punjab floods?	Heavy monsoon rain with western disturbances, high rivers; weak embankments, building near rivers, siltation and late warnings.

★ KEY TAKEAWAYS

- The atmosphere is mainly **nitrogen and oxygen**, with water vapour and dust; it has five layers by temperature and density.
- The troposphere holds the weather; the stratosphere holds the ozone layer.
- **Weather** is short-term; **climate** is the 30-year average.
- Temperature, humidity, precipitation, pressure and wind are the elements of weather and climate.
- India has a **tropical monsoon** climate with four IMD seasons (and six traditional Ṛtus); the SW monsoon brings most rain.
- Climate change from greenhouse gases demands lower carbon footprints, renewable energy and better disaster preparedness, as the Punjab floods show.

CHAPTER 05 · HISTORY

Early Humans and the Beginning of Civilisation

Human history stretches back millions of years, long before kingdoms or writing. This chapter traces our ancestors from Africa through the Palaeolithic, Mesolithic and Neolithic ages, the early Harappan phase, and the Bronze Age civilisations of Mesopotamia, Egypt and China.

Big Questions

- How did humans live before civilisation began?
- How did humans communicate before writing was invented?
- How does archaeology help us understand the past?
- How did early civilisations interact with each other?

Writing and the Study of the Past

The time before writing is known mainly through **archaeological evidence**. Writing appeared at different times in different places. The **Harappan (Sindhu–Sarasvatī / Indus Valley) script**, found on seals and pottery, is pictographic and **still undeciphered** (also called Sindhu lipi). The **cuneiform** of the Sumerians (Mesopotamia) and the **hieroglyphs** of Egypt were contemporary with the Harappans but **have been deciphered**; they mark the start of the historical period about **5,000 years ago**. **Brahmi** was used from about 400 BCE in southern India and the Ganga valley and was formalised by Emperor **Aśhoka** (3rd century BCE).

	Before writing	After writing
Share of human history	More than 99% (about 3 million to 5,000 years ago)	Less than 1% (last 5,000 years)
Main sources	Tools and other artefacts	Material remains **and** written documents
Knowledge of people	Hard to know thoughts and ideas	Literature gives names, events, social, political, cultural life
Measuring time	Only approximate dating	Fairly accurate because documents note dates (coronations, wars)

Why Study Early Human History?

- It shows the long process of **biological evolution** (gradual physical/genetic change from **australopithecines**, 'southern apes', to modern humans, **Homo sapiens**) and **cultural evolution** (adapting to surroundings in the **Quaternary Period**, the last 26 lakh years).
- Humans developed tools and technology; ways of life shifted from hunting and gathering to **agriculture and food production**. The ability to produce **surplus** laid the foundation for civilisation.

▸ Our ancestors

Ancestor	Where / when	Features
Homo habilis ('handy man')	Olduvai Gorge, Tanzania and Kenya; about 2–6 million years ago (as given)	Made chopper stone tools; early toolmaker

Ancestor	Where / when	Features
Homo erectus (upright, bipedal)	Eastern African Rift Valley; about 2 million years ago	Made **handaxes and cleavers**; first hominin to leave Africa, spreading to Asia and Europe
Homo neanderthalensis	Europe and Southwest Asia, until about 40,000 years ago	Made Middle Palaeolithic flake tools
Homo sapiens (modern humans)	Evolved in Africa about 300,000 years ago; exit about 125,000 years ago	Only living human species; developed complex technologies; spread worldwide

- **Hominins** include modern humans and our human-like ancestors. About **3.3 million years ago** one early ancestor made the first stone tools; toolmaking marks the start of 'human behaviour'. Tools are called **extra-corporal limbs** (extensions of the body).
- The earliest settlements were in Africa, Asia and Europe: the **Old World**.
- A **fossil** is the preserved remains, trace or impression of a plant, animal or human of the distant past, formed when remains are buried and slowly turn to stone. Tools and fossils are the main evidence. Archaeologists excavate sites and also **experiment** by making and using similar tools.

Periods of Early Human History

Period	Way of life	Tools / technology
Palaeolithic (Old Stone Age)	Hunter–gatherers; caves and open camps	Simple stone tools; handaxes, cleavers; later scrapers, borers, points; bow and arrow, blades and microblades
Mesolithic	Transition between hunting–gathering and farming; warmer climate; fishing	Microlithic (tiny stone) tools
Neolithic (New Stone Age)	Farming, domestication, permanent villages, pottery, weaving	Polished stone tools; tools for food production
Chalcolithic (Copper–Stone)	Early farming communities	Copper with stone; early metallurgy (extraction and shaping of metals)
Bronze Age	Trade, towns, early civilisations	Bronze (copper + tin)
Iron Age	More advanced societies	Widespread iron smelting; stronger tools and weapons

▸ Palaeolithic hunter–gatherers in India

- India's oldest settlements date to about **2 million years ago**. **Attirampakkam** (Tamil Nadu): about 1.5–1.7 million years. **Isampur** (Karnataka): 1.2 million years. Animal fossils, handaxes and cleavers have been found; these were used to chop meat, dig tubers, scrape skins and break bones for marrow.
- Later: smaller tools (**scrapers, borers, points**) improved hunting by tipping projectiles; then the **bow and arrow**, parallel-sided **blades and microblades** (very sharp glassy rock), small-game hunting, **cave paintings**, body pigments and the first beads of stone, bone and shell. A **burin** (engraver) carved symbols on bone and shell. Microblades were fixed in handles.
- Homo sapiens spread over the world, including Australia and the Americas, between **50,000 and 12,000 years ago**.

▸ Mesolithic hunter–gatherers

About **12,000 years ago** the climate warmed. Forests and grasslands spread over former ice sheets, giving small game, fish and wild grains, so the world saw its first **population explosion**. Microlithic tools helped gather marine and freshwater food; **fishing** was the mainstay. Art flourished and caves and rock shelters were used often. **Bhimbetka** (Madhya Pradesh), a World Heritage Site, has rich cave art.

▸ The Neolithic Revolution

- Hunter–gatherers gained knowledge of seasons and food resources and slowly turned to **food production**: **domestication** of select animals and plants, with new breeds through cultivation and animal husbandry. This is the **Neolithic revolution**.
- Farmers made tools for producing and processing food, made **pottery**, used many raw materials and set up the first **village settlements**, laying the base for the **urban revolution**.
- The shift happened at different times in different regions (West Asia: wheat, barley; India: millets; Ganga plains and Yangtze: rice; North China: millets; West Africa: pearl millet).

▸ Neolithic in the Indian subcontinent

- **Mehrgarh** (Bolan River, present-day Pakistan; foothills of Baluchistan) is the oldest Neolithic site and earliest agricultural village, about **7000 BCE**. People built sun-dried mud-brick houses and granaries, buried the dead in graves, made ornaments from lapis lazuli, carnelian and shell, grew **wheat and barley**, and raised sheep, goats and the zebu humped cattle.
- They were the first to make **copper** objects and became **Chalcolithic** by about **4000 BCE**, laying the basis for the Bronze Age Sindhu–Sarasvatī civilisation around 3500 BCE (as given).
- By **2500 BCE** most of the subcontinent had Neolithic farming communities with herding of cattle, sheep and goats and cultivation of cereals, millets and pulses. A Neolithic site on a granite hill: **Sanganakallu** (about 2000 BCE).

Origins of the Sindhu–Sarasvatī Civilisation

Phase	Approximate date
Neolithic	From 7000 BCE
Chalcolithic	From about 4500/4000 BCE
Bronze Age / Early Harappan	About 3300–2600 BCE
Mature Harappan	2600–1900 BCE
Late Harappan	1900–1300 BCE

- Copper tools in the fertile plains of the Indus and Ghaggar-Sarasvatī basins raised productivity. Radiocarbon dates suggest a pre-Harappan phase at **Bhirrana** and **Kunal** between 7000 and 5500 BCE.
- The **Early Harappan** stage shows continuity in pottery, semi-precious stone beads, shell bangles, terracotta and copper working. Beginnings of **perimeter walls, seals and script** (graffiti on pottery, simple geometric/animal seals at Harappa, Rehman Dheri, Kunal) appear.
- Bead production: Harappa, Kunal (Haryana), Datrana (Gujarat). Pottery was a major craft.
- **Kalibangan** ploughed field (about 5,000 years old) with horizontal and vertical furrows shows **double cropping**, like today's Rabi and Kharif.
- Standard weights: **binary multiples** (1, 2, 4, 8, 16 and so on) for small units and multiples of ten for larger; cubical stone weights eased long-distance trade.

Bronze Age Civilisations Outside India

Four early civilisations arose in **river plains**: Harappan (Sindhu and Sarasvatī), **Mesopotamian** (Euphrates and Tigris, West Asia), **Egyptian** (Nile) and **Chinese** (Huang He / Yellow River). Mesopotamia and the Indus region are close, so contact and trade were strong; Egypt and China show little direct contact with the Harappans.

▸ Mesopotamian civilisation

- **Mesopotamia** = 'land in between' (Greek); today mainly Iraq and Kuwait with parts of Turkey and south-west Iran. The **Fertile Crescent** (Zagros and Taurus foothills from the Mediterranean to the Persian Gulf) had high agricultural potential. Farming began about 12,000 years ago; copper arrived about **4500 BCE**.
- A **city-state** is a sovereign state centred on a city that rules surrounding territory. Four major civilisations from **3500 BCE**: Sumerian, Akkadian, Assyrian, Babylonian.

Civilisation	Date and centre	Key points
Sumerian	From about 3500 BCE; Ur and other cities in Sumer (south)	First city-based civilisation; first dams, canals; mud and burnt bricks; **ziggurat** temple (stepped pyramid temple; city grew around it, served as treasury; temple controlled economy; priests part of ruling class); invented **wheeled cart and sailboat**; base-60 number system (60-minute hour, 60-second minute, 360-degree circle); **cuneiform** from about 3200 BCE
Akkadian	2334 BCE; city of Akkad	Language Akkadian but same cuneiform; **world's first dynastic empire**; King **Sargon** records trade with **Dilmun** (Bahrain), **Magan** (Oman) and **Meluhha** (= Sindhu–Sarasvatī); Harappans traded beads, ivory, timber, gold dust, probably copper. Literature grew (Epic of Gilgamesh)
Assyrian	About 2154 BCE to early 1700 BCE; Assur (north)	Replaced Akkadians; dominance spread west and south
Babylonian	From 1900 BCE; central Mesopotamia	**Hammurabi** (1792 BCE) built a large empire and compiled the **Code of Hammurabi**, a model for later laws (shown receiving the law from Shamash, god of justice). Declined by about 1400 BCE due to **Hittite** attacks (Indo-Europeans in Anatolia, modern Turkey), environmental degradation, farmland pressure, internal problems

DEFINITION

Cuneiform: one of the earliest writing systems; wedge-shaped marks pressed into soft clay tablets with a reed stylus (Latin *cuneus* = wedge). Scribes had high status. Tablets record myths, epics, hymns, law codes, education and farming/craft records. It went out of use by the end of the first millennium BCE.

NOTE

Parallel water management in the Harappan world: **gabarbands** (check dams) in Baluchistan; the water-harvesting tanks of **Dholavira** (Kachchh); the brick dockyard at **Lothal**.

▸ Egyptian civilisation

- Known to Greeks and Romans; **Herodotus** wrote on it in the 5th century BCE. Sources include **papyrus** records (made from the papyrus plant; e.g. Papyrus Ebers with 700+ cures and spells). Libraries stored scrolls in labelled jars; stories like early 'Sindbad' and 'Cinderella' appear.
- **Rosetta Stone**: found in 1799 by Pierre Bouchard, a French army engineer, in three scripts including Greek; **Jean-François Champollion** deciphered hieroglyphs in **1822**.

- Egyptians called their land **Kemet** ('the black'), after the dark flood-soil. The **Nile** flooded each summer, depositing rich mud. City-states emerged about **3000 BCE**. Farmers dug ditches and reservoirs; the need for collective effort led to local government and an administrative class.
- **Calendar:** 365 days (12 months of 30 days + 5), three seasons of four months: Inundation, Peret (growing), Shemu (harvest); based on the rising of Sirius.
- **Pharaohs** were rulers; tombs developed from the **mastaba** into the **pyramid** (e.g. Step Pyramid at Saqqara). Belief in **ka** (spiritual double) led to **mummification** (organs except heart removed, body dried with natron, oiled, wrapped in linen).
- Social pyramid: Pharaoh; officials, nobles, priests; free landholders, artisans, merchants; serfs and slaves.
- Leisure and festivals (statues paraded; sistrum; **Sed festival** for the king's 30th year). Women could own property and run businesses; **Cleopatra** (69–30 BCE) became queen at 18.

Egyptian kingdoms	Dates
Old Kingdom	2686–2181 BCE
Middle Kingdom	2030–1650 BCE
New Kingdom	1570–1069 BCE

▸ Chinese civilisation

- Rivers: **Huang He (Yellow River)** and **Yangtze**; Neolithic cultures about 7000 BCE; copper/bronze about 2000 BCE; urban centres about **1600 BCE**.
- Dynasties: **Shang** (1600–1046 BCE), **Zhou** (1046–256 BCE), **Qin** (221–206 BCE; first imperial dynasty, unified China, name 'China' probably from Qin), **Han** (206 BCE–220 CE). Iron common by about 600 BCE.
- Zhou kings were priest-kings, believed appointed by heaven but removable; officials were examined in archery, horsemanship, calculation, writing and music. China first developed **civil services by public examination** and **paper currency**; money economy by the 5th century BCE.
- Earliest sources: **oracle bones** (animal bones and tortoise shells heated until they cracked and read for prophecy). The script is **logographic** (characters for words).
- Crafts: **jade** objects (ritual/prestige; jade imported), marble carving, bronze weapons, tools and ritual vessels; musical jade pieces. **Silk** known from 4000–3000 BCE; under the Han (2nd century BCE) it became a major trade item, giving the **Silk Route** its name.
- **Great Wall:** walls from 680 BCE by Zhou and others against nomad raids; joined later; repaired until the 17th century CE.
- By 1500 BCE society was stratified: ruling class and nobles, then farmers and labourers.

IMPORTANT

Common features of Bronze Age civilisations: river-valley agriculture, surplus, cities and trade, social hierarchy, administration, writing, craft specialisation and monumental building. They arose independently.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why is the Harappan script not deciphered while cuneiform is?	Harappan inscriptions are short, pictographic and no bilingual key has been found; cuneiform and hieroglyphs have been deciphered.

Likely question	Model answer
Differentiate the periods before and after writing.	Before: mostly tools/artefacts, approximate dating. After: written records too, more accurate dating.
What are biological and cultural evolution?	Biological: physical change from australopithecines to Homo sapiens. Cultural: adapting through tools and technology.
Who was Homo erectus?	Upright ancestor (about 2 million years ago) who made handaxes and cleavers and first left Africa.
How do archaeologists know about our ancestors?	Excavation of tools, bones and fossils, plus experiments with making tools.
Compare Palaeolithic, Mesolithic and Neolithic.	Old Stone Age hunter-gatherers with simple tools; Mesolithic microliths and fishing with a warm climate; Neolithic farming, pottery, villages.
Why is the Neolithic called a revolution?	Domestication of plants and animals changed the way of life, enabling villages and later cities.
Describe Mehrgarh.	Oldest Neolithic site, about 7000 BCE, Bolan River; wheat, barley, zebu, mud-brick houses; first copper by 4000 BCE.
What features of Early Harappan led to the urban civilisation?	Copper tools, pottery, bead-making, perimeter walls, seals, proto-script and trade integration.
What was a ziggurat?	Stepped pyramid temple at the centre of a Sumerian city; economic and treasury centre.
Write on Hammurabi and his code.	Babylonian king (1792 BCE) who compiled laws throughout his empire; a model for later legal systems.
What do we know about Harappan–Mesopotamian trade?	Sargon's records name Meluhha (Harappa), Dilmun, Magan; goods: beads, ivory, timber, gold dust, copper.
How did the Nile help Egypt?	Annual floods left fertile mud (kemet); irrigation, calendar and administration grew from it.
Why did Egyptians mummify bodies?	They believed the ka lived on only if the body was preserved.
Give contributions of the Chinese civilisation.	Oracle bones, jade, silk, Great Wall, civil service examinations and paper currency.

★ KEY TAKEAWAYS

- Early human history is known mostly through archaeology: tools, fossils and cave art.
- Humans evolved in Africa (Homo habilis, Homo erectus, Homo sapiens) and spread across the world.
- Palaeolithic hunter-gatherers → Mesolithic microliths and fishing → Neolithic farming (Mehrgarh, about 7000 BCE) → Chalcolithic copper → Bronze and Iron Ages.
- The Sindhu–Sarasvatī civilisation grew out of Early Harappan farming, crafts and trade.
- Mesopotamia (Sumer, Akkad, Assyria, Babylon), Egypt and China arose in river valleys, with writing, administration and trade.
- Cuneiform and hieroglyphs are deciphered; the Harappan script is not.

CHAPTER 06 · HISTORY

State and Society up to 1000 CE

This chapter follows the Indian subcontinent from clan-based Vedic society to janapadas, mahājanapadas and the great empires of the Mauryas, Guptas and Cholas. It looks at kingship, administration, ethics (dharma), varṇa and jāti, women, religion, education, agriculture, trade and guilds, and shows how deep continuity runs through all of them.

Big Questions

- How did society and state organise themselves before 1000 CE?
- How did states and societies emerge and spread across the subcontinent?
- How did dharma and the chakravarti samrāṭ ideal create unity in values and governance?
- How did social, administrative and occupational groups take shape and come together culturally?

Understanding Early Indian State and Society

DEFINITION

A **society** is a system of social relationships among people who share territory, culture and a sense of belonging; it has families, institutions such as marriage, and is regulated mainly by **custom**. A **State** is an organised political system based on rules and laws, with rights and duties of rulers and subjects, governance mechanisms and institutions for law and order.

From at least the 2nd millennium BCE, literary sources (the **Ṛigveda** first) supplement archaeology, giving a fuller picture of society, politics and ideas. The sweep of the chapter: kin-based polities → janapadas and mahājanapadas → empires (Mauryas, Guptas, Cholas), with continuity in the ideal of a pan-Indian monarch (**aśhvamedha yajña**, **chakravarti samrāṭ**) and in **dharma**.

Period	Development
About 2000 BCE	Kin-based polities of the Ṛigveda (Sapta-Sindhu region)
1000–600 BCE	Janapadas; Iron age; PGW culture
600–300 BCE	Sixteen mahājanapadas; Buddha and Mahāvīra; NBPW
321 BCE	Mauryan Empire founded by Chandragupta Maurya
272–232 BCE	Aśhoka; philosophy of Dhamma
185 BCE	Śhunga empire founded by Puṣhyamitra
300 BCE–300 CE	Sangam literature; early kingdoms of Tamilakam
78 CE / 30–375 CE	Saka Era begins / Kuṣhāṇas
320–550 CE	Guptas
543 CE	Chālukyas founded by Pulakeśhin I
606 CE	Harṣhavardhana annexes Kannauj
712 CE	Arab conquest of Sindh

Period	Development
750–1161 / 753–982 / 730–1036	Pālas / Rāṣhṭrakūṭas / Gurjara-Pratihāras
985–1014 CE	Rājarāja I; Brihadeeshwara Temple, Thanjavur

The Vedic Period

▸ Vedic literature

- The Vedas are the earliest corpus of Indian literature (the Harappan script is undeciphered). They were oral for centuries. The date of the Ṛigveda is debated (5th to 2nd millennium BCE); it was composed in the north-west (**Sapta-Sindhu**: Indus, its five tributaries and the Sarasvatī), where the Sindhu–Sarasvatī civilisation flourished; both knew copper.

Veda	Content
Ṛigveda	Earliest; 1,028 hymns (sūktas) praising deities and reflecting on creation, birth and death
Yajurveda	Performative aspects of yajñas; prose explanations
Sāmaveda	Hymns from the Ṛigveda arranged for musical recitation; basis of the seven svaras
Atharvaveda	Wide range of hymns, some against evil; treatment of physical and mental ailments

Each Veda has four parts: **samhitā** (hymns), **brāhmaṇa** (prose explanation of rituals), **āraṇyaka** (philosophical speculations of forest sages) and **upaniṣhad** (the Self, Ātman, and Ultimate Being, Brāhman). Brāhman is a concept; Brāhmaṇa is a text and also a varṇa.

▸ Political institutions

- Early Vedic society was organised into **janas** (kin-based clans); the Ṛigveda mentions about 30, five of them (Yadu, Turvaśha, Puru, Anu, Druhyu) being the **pañchajana**. 'Bharata' first appears for the **Bharata jana**.
- The **rājā** was a clan chief, leading in war and protecting the clan.

Assembly	Nature
Sabhā	Smaller body of select elites; policy and judicial functions
Samiti	Larger assembly of community members; broader political matters, representing the population
Vidhata	Popular gathering; discussions on warfare and other matters

> **NOTE**
> These assemblies checked the rājā, suggesting an early principle of people's participation in governance.

Early Kingdoms and Republics

- **Janapada** ('where a people first set its foot'): the shift from kinship to territorial identity, between about 1000 and 600 BCE. Control of land, farming and trade routes grew in importance; the Ganga plains replaced the north-west as the cultural centre.
- Archaeology of the first millennium BCE: more settlements in the Ganga region, iron tools, more grains (especially paddy), **PGW** (Painted Grey Ware) pottery and in the middle Ganga (Prayagraj to eastern Bihar) the glossy 'deluxe' **NBPW** (Northern Black Polished Ware).

- **Mahājanapadas** (about 600 BCE–300 CE): bigger units; sixteen are listed (Kamboja, Gandhāra, Matsya, Kuru, Śhūrasena, Pañchāla, Vatsa, Kosala, Kāśhī, Malla, Vṛijji, Magadha, Anga, Avanti, Chedi, Aśhmaka). **Magadha** (Bihar) became the most powerful thanks to location, fertile plains and strong rulers, leading to the **Mauryan Empire**.
- Forms of government: **monarchical** states (**rājyas**) and **republics** (**gaṇas / saṁghas**).
- South: the **Satavāhanas** (Deccan, 2nd century BCE–3rd century CE, nearly 500 years). Further south: **Cholas** (lower Kaveri), **Pandyas** (Tamraparni and Vaigai), **Keralaputras / Cheras**, and **Satiyaputras**, all in Aśhokan inscriptions.

Early Tamil kingdom	Emblem	Region
Chola	Tiger	Fertile rice lands of the Kaveri
Chera	Bow	Kerala
Pandya	Fish	Tamraparni and Vaigai valleys

Known from **Sangam literature** (300 BCE–300 CE); the three rulers were the 'crowned kings' (**Vendar**). A Chera king earned the title **adhirāja**. These early kingdoms should not be confused with the later imperial ones.

Duties and Ideals of the King

- Titles: rājā, mahārāja, samrāṭ. The Arthaśhāstra: 'Only if a king is himself energetically active, do his officers follow him energetically.' The Yajurveda coronation oath: judge the strong and weak fairly, protect the country, do good to the people. The Śhānti Parva of the Mahābhārata guides rulers on ethics and justice.
- The king protected subjects from external threats and internal disorder and dispensed justice (abduction, robbery, theft, adultery); severe punishments were prescribed for grave offences. Kingship was generally hereditary, but kings could also be elected or expelled, so power was not absolute.
- **Pan-Indian awareness:** terms **Jambudvīpa, Bhāratavarṣha, Aśhvamedha and Rājasūya yajña, Prithivi, chakravarti kśhetra**. Aśhoka's edict speaks of changes in the spiritual life of Jambudvīpa; the Arthaśhāstra defines **prithivi** as the area between the Himalayas and the sea = the domain of a universal ruler. The Chera king **Nedunjeral Adan** reached the Himalayas; **Rajendra I** (11th century CE) took the title **Gangaikonda** after conquering the Ganga region.

▸ Council of ministers and the Saptāṅga

DEFINITION

Saptāṅga: Kauṭilya's seven constituents of the state: **king (swāmi), ministers (amātya), territory and population (janapada), forts (durga), treasury (koṣha), army/force (daṇḍa), allies (mitra)**. The state works only when all are integrated. 'One wheel does not move the carriage.'

- The **mantri-pariṣhad** was a small body of elder statesmen advising the king; it generally included the treasurer, chief tax collector, chief legal adviser and commander-in-chief. An Aśhokan inscription shows it could decide in the emperor's absence.
- The **Junagadh Rock Inscription** (near Girnar, Gujarat) bears records of three rulers over about 700 years: **Aśhoka** (Major Rock Edicts in Prakrit, Dhamma), **Rudradaman I** (c. 150 CE; Sanskrit; military achievements and repair of the **Sudarshana Lake**) and **Skandagupta** (5th century CE; restoration of the lake).

How Were Empires Administered?

Level	North	South
Kingdom (rājya)	Divided into provinces: **bhukti**	**Maṇḍala / maṇḍalam**
Division	**Viṣhaya / bhoga**	**Koṭṭam / valanāḍu**
District	**Adhiṣṭhāna / paṭṭana**	**Nāḍu**
Group of villages (tahsil)	**Vithi**	**Paṭṭalā / kūṛram**
Village	Lowest unit; headman **grāmika**	Village

- Satavāhana empire: **āḥāras** (divisions) with **amātyas**. District governors (**pradeśhikas**) did judicial and administrative work, consulting bankers, caravan leaders, artisans and scribes. Cities had governors for revenue and order. About 300–800 CE saw decentralisation.
- **Guptas** kept the old system: mantri heads civil administration; commander-in-chief, general and palace guards chief; new **sāndhivigrahika** (minister of peace and war); **kumārāmātyas** at provincial/local level. The **Damodarpur copper plates** (Kumaragupta I) show a district office of five: head district officer, chief banker, chief caravan trader, chief artisan, chief revenue collector.
- **Pallavas** (c. 275–897 CE): centralised monarchy with local governance; tax-free **Brahmadeya** villages; village assemblies with **variyams** (committees for irrigation, gardens, temples). **Chālukyas** (c. 543–753 CE, Badami): similar; **agrahārams** (Brahmin settlements) at Aihole and Badami became centres of learning.
- **Gurjara-Pratihāras, Pālas and Rāṣhṭrakūṭas** fought the **tri-partite struggle** for **Kannauj**; their administration mixed monarchy with provinces, districts and largely self-reliant villages.
- **Imperial Cholas** (9th–11th century CE): maṇḍalams → valanāḍus → nāḍus → Urs (villages). Village assemblies handled disputes, revenue, land, irrigation, roads and records.

IMPORTANT

Uttaramerur inscription of Parantaka I (10th century CE, Kanchipuram district): describes village elections by the **Kudavolai** ('ballot pot') system. Names of eligible candidates were written on palm leaves in a pot and a young child drew them in public (often at a temple). Winners formed **variyam** committees (tank/irrigation, justice, taxes). Candidates needed 'honest earnings' and 'pure' mind.

Ethics: The Foundation of Law and Life

- Vedas and epics discuss right conduct and the universe. The Bhagavad Gītā stresses knowledge, faith, action and virtue.
- **Samatva** (sameness): all bodies are of the same matter and all consciousness is a manifestation of one supreme consciousness (satya, ṛita). The Mahābhārata repeatedly upholds it across varṇas.
- **Ṛita:** all-pervading cosmic order giving harmony and balance in nature and society (Ṛigveda).
- **Dharma** (not religion): duty, obligation, righteousness, moral conduct, from root *dhri* (to uphold). Bhishma: 'whatever upholds all beings is dharma'; 'the dharma of iron is to sink and of wood to float'. Pāli equivalent: **dhamma**. Aśhoka's edicts promoted dhamma: moral conduct, family respect, non-violence, compassion. Works like Kāmandaka's **Nītisāra** and Bāṇabhaṭṭa's **Kādambarī** (one of the first novels) link ethics and kingship.

Social Structures and Everyday Life

▸ Varṇa and jāti

Varṇa	Expected role
Brāhmaṇa	Study and teach Vedas, perform yajñas, give and receive dāna
Rajanya (Kṣhatriya)	Warfare, protection, justice; also study, sponsor yajñas, donate
Vaiśhya	Agriculture, pastoralism, trade; also study, rituals, charity
Śhūdra	Assist others; in practice farming, herding, trade, crafts

- Earliest mention: **Puruṣhasūkta**, Ṛigveda Book 10. Early Vedic identity depended on many factors (ethnicity, region, village, **gotra**, language, occupation); occupations were flexible. Ṛigveda 9.112.3: 'I am a poet; my father is a physician, my mother grinds corn' shows occupational diversity within a family.
- Varṇa was functional: knowledge ranked highest, then power and wealth. Sutta Nipāta: 'No brāhmaṇa is such by birth but only by his deeds.'
- **Jāti** emerged later through intermarriage, migrating groups becoming **endogamous** (marrying within the group) and territorial differences. Varṇas stayed four; jātis kept multiplying.

Social mobility: rulers came from varied backgrounds (Nandas, Mauryas, Śhuṅgas, Satavāhanas, Vākāṭakas, Guptas, Puṣhyabhūtis). The **Mandsaur inscription** (473 CE) records silk weavers' guild moving from Lata (Gujarat) to Dashapura (Madhya Pradesh), skilled also in archery and astrology. **Karitalai plates** (Jayanātha, late 5th century) show brāhmaṇas as land managers. Sangam works (Tolkappiyam: Arasar, Vanigar, Velar, Antanar; Pattinappalai) show an occupation-based society.

▸ Family, āśhramas and puruṣhārthas

- Smallest unit: **kula** (family) → **grāma** (village) → **viśha** (headed by viśhapati) → **jana** (rājā). Society was **patrilineal**. **Gotra** = patrilineal clan traced to a Vedic ṛiṣhi, guiding marriage.
- Four **āśhramas**: brahmacharya (student), gṛihastha (householder), vānaprastha (forest life), saṁnyāsa (renunciation). Life rites: sixteen **saṁskāras**.
- Four **puruṣhārthas**: **dharma** (righteousness), **artha** (material well-being), **kāma** (desire), **mokṣha** (liberation). Wealth is good when guided by dharma.

▸ Role of women

- Vedic women studied, performed rituals, joined chariot races and the sabhā. Women sages: **Apālā, Viśhvavārā, Ghoṣhā, Lopāmudrā**. Goddesses Uṣha (dawn) and Aditi. Manu-smṛiti 3.56: 'Where women are honoured, there gods rejoice.' Their status later fluctuated.
- Gupta–Vākāṭaka literature shows educated women; **Prabhāvatī Gupta** (daughter of Chandragupta II) ruled as regent in the Vākāṭaka kingdom and issued land grants. Women appear as donors on votive inscriptions.
- Sangam women farmed, herded, spun, fished, made salt and sold garlands; poetesses **Avvaiyar** and **Vennikuyattiyar**. Chola queen **Sembiyan Mahādevī** supported temples.

▸ Religious life and Bhakti

- Vedic gods were forces of nature (sun, rain, fire, dawn) invoked by prayers and **yajñas** (offerings to sacred fire); nature seen as sacred (echoes today in Chhath and Makar Sankranti).
- From the middle of the first millennium BCE came **renunciants** (parivrājaka, bhikṣhu, śhramaṇa); Buddha and Mahāvīra. Vedic worship moved to personal deities Viṣhṇu, Śhiva, Śhakti.

- **Bhakti:** direct devotion open to all regardless of class or gender; prominent from the 6th century in Tamil lands through the **12 Āḻvārs** (Viṣhṇu) and **63 Nāyanmārs** (Śhiva).

The Quest for Knowledge

- Education cultivated truth, patience, humility, self-control and respect for all beings and trained pupils to live by dharma. Subjects: Vedas, grammar, logic, philosophy, ethics, mathematics, science, medicine, astronomy; also music, dance, painting, archery, yoga and meditation.
- **Guru-śhiṣhya paramparā:** the sacred teacher–pupil bond; the **gurukula** at the teacher's home, with the pupil treated as family; disciplined life.
- Higher-learning centres: **Takṣhaśhilā** (planned campus), **Nālandā** (university, 427 CE), **Vikramśhilā** (783–820 CE), **Vallabhī** (480 CE), Odantapurī, Jagaddalā, Vārāṇasī, Kānchīpuram, Śhāradā Pīṭha and others.

Field	Examples
Grammar	Pāṇini's Aṣhṭādhyāyī, Piṅgala's Chhandaśhāstra, Patañjali's Mahābhāṣhya
Law and ethics	Manu-smṛiti, Yājñavalkya, Nārada, Viṣhṇu Smṛiti
Medicine	Charakasamhitā, Suśhrutasamhitā
Sanskrit poetry	Kālidāsa: Raghuvaṃśha, Kumārasambhava
Tamil	Tirukkural, Silappadikāram, Maṇimēkalai, Sangam texts (akam = love, puram = war/public life)

Economy

▸ Agriculture, land revenue and irrigation

- Mauryan state regulated the economy: village land in individual holdings plus common pastures and groves; forests cleared for farming but some protected. Basic land tax: about **one-sixth of produce**. Kauṭilya classified land (cultivated, fallow, high and dry, sown, groves). Crops: rice, pulses, wheat, linseed, mustard, saffron, sugarcane, fruits, vegetables.
- Post-Mauryan: **Milindapañho** (eight stages of farming); cotton on Deccan black soil; **Amarakoṣha** chapter on plants and manure; Sangam texts on Chera jackfruit, pepper, turmeric, ragi and sugarcane.
- **Irrigation:** Puṣhyagupta, Chandragupta Maurya's governor, built a dam on **Sudarshana Lake** (Junagadh record). The Cholas built canals, lakes and tanks; Karikala's **Grand Anicut (Kallanai)** is still used.

▸ Trade, routes and ports

- Magadha traded textiles, gems, coral, pearls, metals and minerals; salt production was state-regulated; the state kept routes safe and checked adulteration.
- By the 6th century BCE two major land routes: **Dakṣhiṇāpatha** and **Uttarāpatha**, linking inland areas with ports. Ports: **Muziris, Kāveripaṭṭinam (Puhār), Arikameḍu, Masulipaṭnam**; coastal and Indian Ocean sea routes. Trade with Rome expanded in the early centuries CE. The Silappadikāram mentions routes from Kanchipuram to Poompuhar and to Kanyakumari.

▸ Guilds and industry

- **Guilds (śhreṇīs)** were associations of people in the same craft or trade; they rose with the mahājanapadas (6th century BCE). They regulated quality and prices, ran guild courts, and acted as **banks, financiers and trustees**. Jātakas mention 18 guilds. Silver **punch-marked coins** circulated.

- **Nāśhik cave inscription** (2nd century CE, Nahapāna's son-in-law Uśhavadāta): donations were deposited with weavers' guilds at fixed interest to maintain monks' caves.
- Textiles (silk, cotton, wool, linen): hubs Mathurā, Kāśhī, Kāmarūpa. Sanchi's southern gateway records the **ivory workers' guild of Vidiśhā** carving the stone sculptures.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What are the four Vedas?	Ṛigveda (hymns), Yajurveda (rituals), Sāmaveda (chanted hymns), Atharvaveda (varied hymns, healing).
Explain sabhā, samiti and vidhata.	Vedic assemblies that checked the rājā: sabhā (elite, policy/justice), samiti (large, community), vidhata (popular gathering).
What is a janapada? How did it differ from a jana?	Territorial unit ('where a people set foot'), replacing kin-based identity with land-based identity.
Why did Magadha become powerful?	Strategic position, fertile plains and strong rulers.
Explain the Saptāṅga theory.	Seven constituents of the state: king, ministers, territory, forts, treasury, army, allies.
What was chakravarti kśhetra?	Domain of a universal ruler: land between the Himalayas and the sea.
Describe administrative divisions in the north and south.	Bhukti/viṣhaya/paṭṭana/vithi/village in the north; maṇḍalam/koṭṭam/nāḍu/kūṛram/village in the south.
What was the Kudavolai system?	Chola village election by drawing palm-leaf names from a pot (Uttaramerur).
What is dharma?	Duty, righteousness and moral conduct that upholds society; not merely religion.
Distinguish varṇa from jāti.	Varṇa: four functional categories, flexible. Jāti: later birth-based groups, unlimited in number.
What were the āśhramas and puruṣhārthas?	Four life stages and four life goals (dharma, artha, kāma, mokṣha).
What do we learn about women in early India?	Scholars, rishikas, regents (Prabhāvatī Gupta), Sangam workers and poets, donors.
Who were the Āḻvārs and Nāyanmārs?	Tamil Bhakti saint-poets devoted to Viṣhṇu (12) and Śhiva (63).
Describe the gurukula system.	Pupils lived with the guru, learned holistic subjects and values, in a disciplined sacred relationship.
What was the role of guilds?	Regulated crafts and trade, quality and prices; gave credit and held endowments as banks.
Why were irrigation works recorded in inscriptions?	Agriculture was the main revenue source, so irrigation shows state care (Sudarshana Lake, Kallanai).

★ KEY TAKEAWAYS

- Society evolved from kin-based janas to territorial janapadas, mahājanapadas and empires; Magadha led to the Mauryas, followed by the Guptas and Cholas.
- The king protected, gave justice and was advised by a council; assemblies and Saptāṅga limited and structured power.
- Administration ran through provinces, districts and largely autonomous villages (Uttaramerur elections).
- **Dharma** and ethics guided rulers; the pan-Indian ideal of the chakravarti samrāṭ recurs across centuries.
- Varṇa began as functional and flexible; jāti grew later; women, Bhakti and education show a rich, evolving society.
- Agriculture, irrigation, trade routes, ports and guilds powered the economy.

CHAPTER 07 · POLITY

Democracy

Democracy is the form of government in which power rests with the citizens, who choose their representatives in free and fair elections. This chapter traces the roots of democratic ideas in India, the constitutional principles that sustain them, the forms of democracy in the world, how India's democracy lives at the grassroots, and the challenges it faces.

Big Questions

- How has the idea of democracy evolved and shaped our understanding today?
- What forms of democracy exist and how are they practised in India and elsewhere?
- What are the essential features of democracy beyond government institutions?
- What challenges does Indian democracy face?

India: The World's Largest Democracy

- India has a population of over **140 crore** and over **96.8 crore** registered voters (2024): the largest participatory democracy.
- The **Constitution of India** was **adopted on 26 November 1949** and **came into force on 26 January 1950**. It guarantees and protects Fundamental Rights and bars discrimination on grounds of religion, race, caste, sex or place of birth; it encourages participation at national, state and local levels.
- Democratic ethos did not arise suddenly; it grew over a long historical process.

Tracing Democratic Traditions from Early Times

- **Vedic assemblies** (sabhā, samiti, vidhata) and early **republics (gaṇas/saṁghas)** show collective decision-making; the king worked in consultation with assemblies, ministers and officers.
- The **Aikyamatya Sūktam** (Ṛigveda 10.191.3) stresses one counsel, one assembly, one mind: consultation, consensus and shared responsibility.
- The **Bauddha Saṁghas** (Buddha's monastic community) encouraged debate, chose their leader and decided by voting.
- Foreign invasions and, by the 19th century, **British colonisation** limited participation, but the **freedom struggle** revived democratic ideas.
- The **Constituent Assembly** was formed in **1946** and took **2 years, 11 months and 18 days** to draft the world's longest written constitution. **Dr B.R. Ambedkar** chaired the Drafting Committee. Its debates (CAD) wanted a flexible, dynamic constitution; **Article 368** allows amendments without compromising basic values (liberty, equality, justice, fraternity).

Principles of Democracy

Principle	Meaning and constitutional provisions
Popular sovereignty	Ultimate power lies with the people; the state derives authority from them. Citizens aged **18 and above** vote by **secret ballot**: **Universal Adult Franchise**. (*Sovereignty* from Latin *superanus* 'supreme': the supreme, independent legal authority of a state over its territory and citizens.)

Principle	Meaning and constitutional provisions
Rule of law	Equality before law and equal protection; nobody is above the law; disputes settled in courts; no punishment except by **procedure established by law** (**Article 21**: no one deprived of life or personal liberty except by this procedure). *Equality before law* = same treatment irrespective of rank; *equal protection* = like persons in like circumstances treated alike, without arbitrary discrimination
Fundamental Rights	Protect freedom and dignity; enforceable in courts; reasonable restrictions possible
Separation of powers	Legislature makes laws, executive implements, judiciary interprets, preventing concentration of power and creating checks and balances; the judiciary can declare a law unconstitutional and review amendments. **PIL** (Public Interest Litigation) widens access to justice
Accountability and transparency	Government answerable to citizens through elections, debate, civil society; **RTI Act, 2005**; Question Hour; audit of accounts
Multi-party system	Many parties represent diverse interests; the party/coalition with more than 50% seats forms the government, the largest opposing group becomes the opposition; governed by the **Representation of the People Act, 1951**
Protection of vulnerable groups	Equitable resource distribution. **Article 46**: State shall promote with special care the educational and economic interests of weaker sections, especially SCs and STs, and protect them from social injustice and exploitation

Fundamental Right	Articles
Right to Equality	14–18
Right to Freedom	19–22
Right against Exploitation	23–24
Right to Freedom of Religion	25–28
Cultural and Educational Rights	29–30
Right to Constitutional Remedies	32 (and 226 for High Courts)

> **NOTE**
>
> **Right to Education (Article 21A)**, added in 2009, gives free and compulsory education to children aged 6 to 14. The Constitution also contains Fundamental Duties and Directive Principles of State Policy.

▸ Institutions and the values they promote

Process	Institution	Values
Legislative	Parliament, State legislatures, local bodies	Representation, deliberation, dissent
Electoral	Election Commission of India	Participation, equality
Judicial	Courts	Rule of law, equality, justice
Participatory	Media, civil society	Debate, discussion, freedom of expression

Process	Institution	Values
Accountability and transparency	CAG, Central Information Commission, Lokpal, Central Vigilance Commission	Transparency
Decentralisation	Rural and urban local bodies	People's participation at grassroots
Federalism	Division of powers between Centre and states by a written constitution	Devolution of power

DEFINITION

Civil society: voluntary groups, NGOs and community associations functioning within society. The **media** is called the **fourth pillar of democracy** because it informs people and voices public issues. In Question Hour, members question the government.

Types of Democracy

	Direct democracy	Representative (indirect) democracy
Who decides	Citizens take part directly in most decisions	People elect representatives; they do not govern directly
Suitability	Hard in large countries	Suits large countries; periodic elections; government accountable
Example	Switzerland	India

	Parliamentary	Presidential
Executive	Part of the legislature and accountable to it	Independent of the legislature
Head	People elect legislature, not executive	President elected by people and accountable to them
Examples	India, Canada	USA

Democracy has no single model: freedom, equality, participation, rule of law, rights and duties are common, but forms differ. (The school-council example: Team A direct; Team B representative; Team C presidential; Team D has elections but concentrated power and weak opposition, so it is not truly democratic.)

Country	Type	Executive / head	Key features
India	Parliamentary representative	PM (head of government), President (head of state)	Multi-party; written constitution; federal (Union, State, Concurrent lists; residual powers with Union)
Canada	Parliamentary	PM; Governor-General appointed by the Monarch on PM's advice	Federalism, multi-party
United Kingdom	Parliamentary constitutional monarchy	Monarch (head of state), PM	Unwritten constitution, parliamentary sovereignty
Switzerland	Direct democracy	Federal Council; President is head of government	Written constitution, many parties

Country	Type	Executive / head	Key features
USA	Presidential	President is head of government	Written constitution, two major parties

DEFINITION

Federalism: a system in which power and responsibilities are shared between the Union and the state governments.

Democracy in Practice: India's Living Democracy

- An MP represents on average about **25 lakh** people (2019). Elections are held in **22 scheduled languages** and more; **election symbols** let non-readers vote. More than **one million polling stations**, even in mountains, forests, deserts and islands (sometimes for one voter). Over **2,800** registered political parties.
- **Participatory democracy** stresses citizens' involvement beyond voting. India has a three-tier system: Union, State, Local. Autonomous District Councils (ADCs) in certain north-eastern areas have legislative and judicial powers to preserve tribal customs; the **PESA Act, 1996** empowers the **Gram Sabha** in tribal areas of other states.

▸ Case study: democracy at the grassroots

Panchayat	Achievements
Jethipura Gram Panchayat (Idar block, Sabarkantha, Gujarat)	Won the **Nanaji Deshmukh Rashtriya Gaurav Gram Sabha Puraskar** (2019): effective Gram Sabhas, record keeping, special Gram Sabhas, Mahila Sabhas, adequate **quorum** and participation of SCs, STs and women; action on education, sanitation, welfare
South Manubankul Gram Panchayat (Rupaichari, Tripura)	Nationally recognised as a **women-friendly panchayat** by the Ministry of Panchayati Raj: women in decision-making, health, sanitation, livelihoods, self-help groups

Democratic habits also appear in cooperatives (members share profits and responsibilities), trade unions, civil society and community institutions.

Women and the Right to Vote

- Women got full voting rights in **Britain in 1928** and the **USA in 1920** after long struggles; in India the 1950 Constitution gave **universal adult franchise** to men and women alike, without a struggle. Yet social prejudice still limits women's representation.
- **Article 243(d)**: not less than one-third of seats in every Panchayat reserved for women; **21 States and 2 UTs** provide 50%. **Article 243(t)**: one-third in every Municipality; as of 2023, **17 States and 1 UT** provide 50% in urban local bodies.

NOTE

RTI Act, 2005 came through Parliament after citizens, journalists and civil society raised transparency concerns, parliamentary committees and expert groups examined them, and elected representatives added safeguards for national security and administrative efficiency. It lets citizens seek information from public authorities.

Challenges to Democratic Practices in India

- Rights come with responsibilities. Damaging public property, spreading misinformation or ignoring public issues weakens democracy.
- Challenges: **illiteracy, misinformation and fake news** (especially on social media), **inequality, poverty, regionalism, gender inequality, social discrimination** and gaps in implementing laws, which reduce trust in institutions.

▸ The Emergency (1975–77)

- Early 1970s: unemployment, inflation and allegations of misgovernance under **Indira Gandhi** led to protests. In **June 1975** a National Emergency was declared on grounds of internal disturbance (**Articles 352, 356, 360**: National Emergency, President's Rule, Financial Emergency).
- Most Fundamental Rights were suspended, press censored, leaders and activists arrested. Mass movements led by **Jayaprakash Narayan (Lok Nayak)** mobilised people, especially in Bihar and Gujarat.
- Lifted in **1977**; the ruling government lost the general election, showing the strength of democracy and the need for constitutional safeguards, civil liberties and citizen vigilance.

▸ Democracy and you

Be well-informed and media literate; follow Fundamental Duties; understand the Constitution; join **NSS, NCC, Bharat Scouts and Guides**; show leadership at school and in the community. **Civic responsibility** includes respecting others' rights, using social media responsibly, obeying laws, valuing diversity and supporting unity.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is democracy?	Government in which power rests with the people, who elect representatives in free and fair elections.
Show that democratic ideas are rooted in ancient India.	Sabhā, samiti, vidhata, gaṇas/saṁghas, Aikyamatya Sūktam and the Bauddha Saṁgha's voting.
How long did the Constituent Assembly take to draft the Constitution?	2 years, 11 months and 18 days; formed in 1946.
What is popular sovereignty and Universal Adult Franchise?	Power derives from the people; every citizen of 18+ may vote by secret ballot.
Explain the rule of law.	Equal law for all, disputes in courts, no punishment without procedure established by law (Article 21).
List the six Fundamental Rights with articles.	Equality 14–18; Freedom 19–22; Against Exploitation 23–24; Religion 25–28; Cultural and Educational 29–30; Constitutional Remedies 32.
Explain separation of powers.	Legislature makes, executive implements, judiciary interprets laws, giving checks and balances.
How is accountability ensured?	Elections, debates, Question Hour, audit (CAG), RTI Act 2005, CIC, Lokpal, CVC.
Distinguish direct and representative democracy.	Direct: citizens decide (Switzerland). Representative: elected representatives decide (India).

Likely question	Model answer
Differentiate parliamentary and presidential democracy.	Parliamentary: executive accountable to the legislature (India, Canada). Presidential: separately elected President (USA).
Why is India called the largest democracy?	Over 96.8 crore voters, huge polling network, many languages, 2,800+ parties.
What reservation exists for women in local bodies?	Not less than one-third (Art. 243(d), 243(t)); many states give 50%.
What were the causes and effects of the Emergency?	Economic distress and protests led to its imposition in 1975; rights and press curbed; lifted in 1977 and the government lost the election.
Name challenges to Indian democracy.	Illiteracy, fake news, inequality, poverty, regionalism, gender and social discrimination, weak implementation.
Why is media called the fourth pillar?	It informs citizens and voices public issues, holding government accountable.

★ KEY TAKEAWAYS

- Democracy means self-governance by participation, not just elections.
- India's democratic roots lie in Vedic assemblies, republics and Buddhist Saṁghas; the Constitution (1950) gives it modern form.
- Key principles: popular sovereignty, rule of law, Fundamental Rights, separation of powers, accountability, multi-party system, protection of the vulnerable.
- Democracy has many forms: direct or representative, parliamentary or presidential.
- Local bodies, women's reservation and RTI deepen grassroots democracy.
- Fake news, inequality and illiteracy are challenges; the Emergency showed both vulnerability and resilience; informed, responsible citizens sustain democracy.

CHAPTER 08 · POLITY

Elections

Elections are the heart of democracy: they let citizens choose who will decide on their behalf and hold those leaders to account. This chapter explains why elections matter, how seats are won under different electoral systems, the laws and institutions that run Indian elections, the work of the Election Commission, the role of political parties and the challenges to free and fair polls.

Big Questions

- Why are elections essential to a democratic system?
- How do electoral systems and rules shape representation and fairness?
- What role do institutions and laws play in ensuring free and fair elections?
- What challenges do elections face and how do they affect democracy?

Why Do Elections Matter?

	Direct elections	Indirect elections
Meaning	Citizens vote directly for their representatives	Citizens vote for representatives who then choose the leaders
In India	**Lok Sabha** (Centre), **Vidhan Sabha** (state), local bodies such as Panchayats and municipal corporations; every five years	**President, Vice President, Rajya Sabha** members

- A class representative (like Gurmat in Grade 8) cannot continue for ever without a fresh election; similarly a government cannot stay beyond its term without a fresh mandate, and choice is meaningless if only one party contests. Hence **periodic elections** are essential.
- Elected representatives are **accountable** to the people; the vote is the main means of making them so. Elections ensure **representation, accountability, legitimacy and equality**.

NOTE

Psephology is the scientific study of elections, from Greek *psēphos* ('pebble', used for voting in ancient Greece) + *logy*.

Inside India's Election Machinery

Free, fair and transparent elections in a country with millions of voters and thousands of constituencies need laws, coordinated machinery at national, state and local levels, and stakeholders: the **Election Commission of India (ECI)**, political parties, civil society, media and voters.

▸ The electoral system

An electoral system is the method of converting votes into seats. The Constitution makers chose the **plurality / First-Past-The-Post (FPTP)** system.

Body	System
Lok Sabha and Vidhan Sabha	FPTP

Body	System
Rajya Sabha, President, Vice President	Proportional representation (by single transferable vote)
Vidhan Parishad (Legislative Council)	Proportional representation by single transferable vote

- **Vidhan Sabhas** are directly elected and make laws for states and some Union Territories. Six states have **two houses** (bicameral): Andhra Pradesh, Bihar, Karnataka, Maharashtra, Telangana and Uttar Pradesh. The **Vidhan Parishad** has members elected by MLAs and local bodies, graduates and teachers, plus some nominated by the **Governor** for art, science, social service, literature and cooperatives.

Feature	FPTP	Majority (two-round) system	Proportional representation (PR)
How voting works	Vote for one candidate in a constituency; most votes wins, even below 50%	Candidate needs **more than 50%**; if none, the top two compete again in a second round	Voters vote for a **party**; seats allotted in proportion to votes
Imaginary-country example (100 seats / 25 / list)	Party A 40%, B 35%, C 25%: A wins and forms the government	Round 1: X 45%, Y 40%, Z 15% (no one over 50%); Round 2: Y wins with 55%	Party M 40, N 35, O 15, P 10 seats; power shared proportionately

FORMULA

Single transferable vote: Quota = (Total valid votes ÷ (Seats to be filled + 1)) + 1. Voters mark preferences; first preferences are counted and those reaching the quota are elected; the lowest candidate is eliminated and votes move to the next preference; repeat until all seats are filled.

▸ The laws

Law	Covers
Representation of the People Act, 1950	Allocation of seats, **delimitation** of constituencies, preparation and revision of **electoral rolls**; right of every citizen above 18 to vote without discrimination
Representation of the People Act, 1951	All other aspects of conduct: nomination, campaigns, voting, resolution of disputes; lists **electoral offences and corrupt practices**
Presidential and Vice-Presidential Elections Act, 1952	Elections to the offices of President and Vice President

Corrupt practices (RPA, 1951): gifts, offers or promises to induce someone to contest, withdraw, vote or not vote; appeals for votes on grounds of religion, caste, race, community or language; taking help for the candidate from government personnel such as gazetted officers, judges and magistrates, armed forces, police and excise officers.

▸ Delimitation Commission

- **Delimitation** is fixing the number of seats and territorial boundaries of constituencies for the Lok Sabha and Vidhan Sabhas so that seats-to-population ratios are as equal as possible. Without it one MP might represent five lakh people and another twenty-five lakh.
- **Article 82** requires a Delimitation Commission. There have been four: **1952, 1963, 1973 and 2002**.

Election Commission of India (ECI)

- An **autonomous, permanent constitutional body**, established **25 January 1950**; **Articles 324 to 329** give its powers. It has superintendence, direction and control of elections to the Lok Sabha, Rajya Sabha, Vidhan Sabha, Vidhan Parishad, President and Vice President.
- In states the **Chief Electoral Officer** (appointed by the ECI from civil servants in consultation with the state government) supervises; District Magistrates, SDMs and Tahsildars act as District Election Officers, Returning Officers and Electoral Registration Officers.
- **25 January** is celebrated as **National Voters' Day** with the voters' pledge to uphold democratic traditions, free, fair and peaceful elections and to vote fearlessly, without influence of religion, race, caste, community, language or inducement.

Function of ECI	Details
Prepares the electoral roll	Enumerators visit households; rolls are made constituency-wise and polling-station-wise; only listed persons may vote. **Special Intensive Revision (SIR)**: updates, verifies and corrects rolls so that no eligible citizen (especially new 18-year-olds) is left out and no ineligible person is included; removes names for death, change of residence, duplicate entry, untraceable voters; claims and objections are invited and settled before the final roll
Fixes the election schedule	Acts when the five-year term ends or the House is dissolved early; considers weather, agricultural cycle, school and university exams, festivals
Registers parties and allots symbols	Only registered parties may contest; independents can also contest. Insists on inner-party democracy (periodic organisational elections); classifies parties as national, state/regional or registered-unrecognised; allots symbols; acts as a **quasi-judicial** body in disputes over recognition and symbols
Ensures free and fair elections	Inclusive, impartial and trustworthy process; Model Code of Conduct; EVMs and VVPAT; awareness campaigns

DEFINITION

ETPBS (Electronically Transmitted Postal Ballot System): lets **service voters** receive and cast postal ballots electronically from outside their constituency.

▸ Inclusion and technology

- Focus on persons with disabilities (PwDs), senior citizens, transgender persons and Particularly Vulnerable Tribal Groups (PVTGs). **Voting from home** for those above **85 years** and PwDs with **40% benchmark disability** was extended nationwide for the first time in the **2024 General Elections**. EVMs with **Braille**; ramps, wheelchairs and volunteers.

App / system	Purpose
Voter Helpline App	Registration, transposition, verification, information
Saksham App	Services for PwDs: registration, finding the polling station
cVIGIL	Report Model Code of Conduct violations in minutes; flying squads act using auto-location

App / system	Purpose
Suvidha	For candidates and parties: nomination forms and permissions (rallies, vehicles, loudspeakers) with real-time status
Sugam	Managing vehicles requisitioned for election work
ERONET	Seamless processing of forms and electoral rolls (e-Rolls)
ETPBS	Electronic postal ballots for service voters

NOTE

International Election Visitors' Programme (IEVP): ECI's flagship for cooperation with election bodies of other countries; MoUs with **28 EMBs** and three organisations: **IFES, International IDEA and the United Nations**. India has over 96.8 crore eligible voters.

Political Parties

Parties present policies and vision, help voters learn about candidates and issues, choose the election agenda and work at the grassroots. They:

- organise public opinion and campaign on issues;
- offer voters meaningful choices;
- enable informed decisions through rallies, manifestos and debate;
- contest elections, form the government (winning party) or sit as the opposition, formulate policies and ensure accountability in debates and committees.

India has a **multi-party system** reflecting its diversity.

▸ Defection and the Anti-Defection Law

DEFINITION

Defection: abandoning the party on whose ticket one was elected, by joining another party or acting against it (e.g. in voting). It may be seen as opportunism (against party discipline and stability) or as following conscience.

- **Anti-Defection Law:** passed in **1985** by the **52nd Constitutional Amendment Act** to give stable governments. A member of Parliament or a State Assembly who voluntarily gives up party membership or votes against the party **whip** can be disqualified; the **Speaker or Chairman** decides.

▸ Recognition of parties

Party type	Criteria (any one)
National party	6% of valid votes in any four or more states in a Lok Sabha/Assembly election **plus** four Lok Sabha seats; **or** 2% of Lok Sabha seats from at least three states; **or** recognised as a state party in at least four states
State party	6% of valid votes in the state in a general election to the Assembly **plus** two Assembly seats; **or** 6% of valid votes in a Lok Sabha election from the state **plus** one Lok Sabha seat; **or** 3% of Assembly seats (or three seats, whichever is more); **or** one Lok Sabha seat for every 25 seats (or fraction) allotted to the state; **or** 8% of valid votes in the state in a Lok Sabha or Assembly election

Party type	Criteria (any one)
Registered unrecognised (RUPP)	Parties that have not secured enough votes, or have never contested since registration

NOTE

Alliance politics: 1967 ended single-party dominance and began an era of alliances. In **1977** the **Janata Party** formed the first coalition government at the Centre after the Emergency (1975–77). Since the late 1990s the **NDA** (led by the BJP) and the **UPA** (led by the Congress; dissolved in 2023 and reconstituted as the **INDIA** alliance) are the major alliances.

Challenges to Free and Fair Elections

- Conducting elections for over 96.8 crore voters (2024) across thousands of polling stations and hundreds of parties is difficult. Challenges include **misinformation, fake news, intimidation**, money power and others.
- Responses: **RPA 1950 and 1951**, **Model Code of Conduct**, **EVMs**, **VVPAT** (Voter Verifiable Paper Audit Trail), voter-awareness campaigns, vigilance apps. Active citizen participation makes elections more representative and democracy stronger.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why are periodic elections important?	They make rulers accountable, give legitimacy and equal voice, and allow people to change governments.
Distinguish direct from indirect elections with examples.	Direct: Lok Sabha, Vidhan Sabha, local bodies. Indirect: President, Vice President, Rajya Sabha.
What is psephology?	Scientific study of elections, from Greek psephos (pebble).
Explain the FPTP system.	Candidate with the most votes in a constituency wins even without 50%. Used for Lok Sabha and Vidhan Sabha.
How does proportional representation differ from FPTP?	PR gives parties seats in proportion to votes; used for Rajya Sabha, President, Vice President, Vidhan Parishad.
What does the Representation of the People Act 1950 provide? And 1951?	1950: seats, delimitation, electoral rolls, adult franchise. 1951: conduct of elections, offences, disputes.
What is delimitation? Who conducts it?	Fixing seats and constituency boundaries to equalise representation; Delimitation Commission under Article 82 (1952, 1963, 1973, 2002).
Write a note on the Election Commission of India.	Autonomous constitutional body (25 Jan 1950, Articles 324–329) that prepares rolls, schedules elections, registers parties, allots symbols and ensures fairness.
What is SIR?	Special Intensive Revision updates and corrects electoral rolls so no eligible voter is left out and no ineligible one is included.
Name ECI initiatives for inclusion.	Voting from home for 85+ and 40% PwDs (2024), Braille EVMs, Saksham, ETPBS, Voter Helpline, cVIGIL.

Likely question	Model answer
What is cVIGIL?	App to report Model Code of Conduct violations; flying squads respond using auto-location.
What is the Anti-Defection Law?	52nd Amendment (1985): members who leave their party or defy the whip may be disqualified by the Speaker or Chairman.
State two criteria for recognition as a national party.	6% votes in four states plus four Lok Sabha seats; or 2% Lok Sabha seats from three states; or state-party status in four states.
What roles do political parties play in elections?	Organise opinion, offer choices, campaign, form government or opposition, make policy.
What are the challenges to free and fair elections and remedies?	Misinformation, fake news, intimidation, money power; remedies: RPA, Model Code of Conduct, EVM, VVPAT, awareness.

★ KEY TAKEAWAYS

- Elections are the soul of democracy: they ensure accountability, representation, legitimacy and equality.
- India uses **FPTP** for the Lok Sabha and Vidhan Sabha, and proportional representation (single transferable vote) for the Rajya Sabha, President, Vice President and Vidhan Parishad.
- The **RPA 1950** and **1951** and the **Delimitation Commission** structure fair elections.
- The **ECI** (Articles 324–329) prepares rolls, schedules polls, registers parties, allots symbols and promotes inclusion through technology.
- Political parties inform voters and form governments or opposition; the Anti-Defection Law (1985) gives stability.
- Misinformation, fake news and intimidation challenge fair elections; vigilant citizens and the ECI address them.